

REED–SOLOMON MCA THRESHOLDS

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ABSTRACT. We determine exact finite thresholds in broad regimes and rigorous asymptotic threshold compilers for affine support-wise mutual correlated agreement of Reed–Solomon codes. If C has parameters $[n, k]$, Γ is a nonempty challenge set, and the integer $0 \leq r \leq n - k - 1$ satisfies $(n - r)^2 \geq n(k + r)$, then $B_{C, \Gamma}^{\text{MCA}}(n - r) = \min\{|\Gamma|, r + 1\}$; the same upper bound holds for every linear MDS code. An exact CA–sparse decomposition, together with the cited lossless unique-decoding theorem, extends the staircase asymptotically through every fixed relative radius below half the minimum distance, while syndrome–secant geometry and locator–prefix pole lines give independent finite upper and lower mechanisms. We obtain exact target compilers, four certified multiplicative rows at target 2^{-128} , a shallow-prefix asymptotic exponent, and the formula $\delta^* = 1 - \rho - g + o(1)$ whenever matching certificates meet at agreement $k + 1 + gn + o(n)$. Generalized Reed–Solomon and explicitly diagonalized circle codes inherit the finite conclusions, but no unrestricted near-capacity smooth/circle claim is made.

1. INTRODUCTION AND MAIN RESULTS

By a finite Reed–Solomon *row* we mean data $(\mathbb{F}, D, k)$, where $\mathbb{F}$ is a finite field, $D \subseteq \mathbb{F}$ is an *evaluation domain* of size n , and $1 \leq k < n$. The associated code $C = \text{RS}_{\mathbb{F}}(D, k) \subseteq \mathbb{F}^D$ consists of the evaluations on D of polynomials of degree less than k . Thus a support is simply a subset of the coordinate set D . A word $u \in \mathbb{F}^D$ is *explained on a support* $S \subseteq D$ if its restriction to S agrees with one such degree-less-than- k polynomial. When the evaluation points lie in a designated coefficient subfield, we write $D \subseteq \mathbb{B} \subseteq \mathbb{F}$; $\mathbb{B}$ controls locator coefficients and domain structure, while $\mathbb{F}$ controls codeword coefficients and slopes. Throughout, $h(x) = -x \log x - (1 - x) \log(1 - x)$ denotes binary entropy in natural units, and $H_2(x) = h(x)/\log 2$ denotes binary entropy in bits, with the usual convention $0 \log 0 = 0$.

A received pair $r = (r_0, r_1) \in (\mathbb{F}^D)^2$ determines the *received line* $\{r_0 + \gamma r_1 : \gamma \in \mathbb{F}\}$. The pair is simultaneously explained on S if each of the two received words has an explanation there, possibly by different polynomials. A finite slope γ is *support-wise MCA-bad at agreement* a if some support $S \subseteq D$, $|S| \geq a$, explains $r_0 + \gamma r_1$ but does not simultaneously explain the pair. The same support is used in both tests; this is the support-wise convention. A tuple (γ, S, h) , where $\deg h < k$ and $h|_S = (r_0 + \gamma r_1)|_S$, is a *witness* to that bad slope when the pair is not simultaneously explained on S , or equivalently is *support-wise nontrivial* there. Finally, $B_C^{\text{MCA}}(a)$ is the maximum, over all received pairs, of the number of distinct bad slopes. It counts slopes, not witnesses or supports.

This is exactly the affine-line definition used by the Proximity Prize [PP26] and by Arnon–Boneh–Fenzi [ABF26, Definition 4.3]: for a real radius δ , a witnessing support has size at least $(1 - \delta)n$. Since support size is integral, put

$$r(\delta) = \lfloor \delta n \rfloor, \quad a(\delta) = n - r(\delta).$$

Thus the error is a right-continuous integer staircase. In particular, a statement about a “largest” real radius needs an endpoint convention; in the exact rows below the safe set is half open and has a supremum but no largest element.

Here is the asymptotic formulation used throughout. If $\emptyset \neq \Gamma \subseteq \mathbb{F}$ is the *challenge set*, meaning the set of slopes allowed in the test, let $B_{C, \Gamma}^{\text{MCA}}(a)$ denote the same maximum with

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$\gamma \in \Gamma$, and at the radius grid point $\delta = 1 - a/n$ put

$$(1.1) \quad e_{\text{MCA}, \Gamma}(C, \delta) = \frac{B_{C, \Gamma}^{\text{MCA}}(a)}{|\Gamma|}.$$

For the standard full-field challenge we abbreviate

$$e_{\text{MCA}}(C, 1 - a/n) = \frac{B_C^{\text{MCA}}(a)}{|\mathbb{F}|}, \quad \delta_{C, \text{sup}}^*(\varepsilon) = \sup\{0 \leq \delta < 1 - k/n : e_{\text{MCA}}(C, \delta) \leq \varepsilon\},$$

where the error at an arbitrary real radius is evaluated at $a(\delta) = n - \lfloor \delta n \rfloor$. We set this supremum to zero when the safe set is empty. We separately write

$$\delta_{C, \text{grid}}^*(\varepsilon) = \max \left\{ \frac{r}{n} : 0 \leq r \leq n - k - 1, e_{\text{MCA}}(C, r/n) \leq \varepsilon \right\}$$

when this set is nonempty. If the first unsafe integer radius is r_0 , then $\delta_{C, \text{grid}}^* = (r_0 - 1)/n$ while $\delta_{C, \text{sup}}^* = r_0/n$, and the latter endpoint is unsafe. Taking $\Gamma = \mathbb{F}$ in (1.1) gives exactly this customary formulation. By a *row sequence* we mean a family $C_n = \text{RS}_{\mathbb{F}_n}(D_n, k_n)$ indexed so that $|D_n| = n$; the fields, domains, and dimensions may all vary with n . Every claimed upper bound is uniform over received pairs, whereas a lower construction need only exhibit the stated received pair. For such a sequence, challenge sets Γ_n , and targets $0 \leq \varepsilon_n \leq 1$, define

$$(1.2) \quad \begin{aligned} B_n^* &= \lfloor \varepsilon_n |\Gamma_n| \rfloor, \\ a_n^* &= \min\{a \in \{k_n + 1, \dots, n\} : B_{C_n, \Gamma_n}^{\text{MCA}}(a) \leq B_n^*\}, \quad \delta_n^* = 1 - \frac{a_n^*}{n}, \end{aligned}$$

whenever the set defining a_n^* is nonempty. Here δ_n^* denotes the largest safe *grid* radius; the corresponding continuous supremum is one grid step to the right whenever $a_n^* - 1$ is the first unsafe agreement. Thus the asymptotic RS-MCA problem is to determine δ_n^* , or equivalently the first *safe agreement*: agreement a is safe when its bad-slope numerator is at most B_n^* , and unsafe otherwise. Since the numerator is nonincreasing in a , the resulting safe/unsafe boundary is the *MCA frontier*. This formulation retains the target and challenge size; replacing them by a zero-exponent slogan loses real information when the target is small.

Position relative to current work. A proximity-gap theorem says, roughly, that an affine line which is not entirely close to the code contains only a controlled number of close points. For a rate- ρ Reed-Solomon code, the unconditional positive theory for arbitrary prescribed domains reaches the Johnson radius $1 - \sqrt{\rho}$, with recent refinements and MCA conversions [BCIKS20, BCHKS25, BCGM25, Hab25]. Capacity-level positive results concern folded, multiplicity, subspace-design, or random-evaluation models [GG25, GGSW26], while fixed smooth subgroups have explicit near-capacity obstructions [CS25, KKH26, Kam26]. Two priority distinctions are essential. First, the two-radius theorem [BCHKS25, Thm. 1.3], the below-half equality of the global CA and MCA errors [ACFY25, Lem. 4.10] (restated as [ABF26, Lem. 4.6]), and the matching perturbation already imply the exact $r + 1$ numerator for constant-rate RS rows through almost half their minimum distance. The exact CA/sparse decomposition proved here is a self-contained refinement that also retains restricted challenge sets. Thus neither the constant-rate consequence of the quadratic theorem nor the four large Proth-row applications below is claimed as a new consequence. Second, the syndrome-space viewpoint also occurs independently in [YZ26].

We are unaware of a prior finite MDS statement of the following form: the quadratic overlap argument applies without a large-length hypothesis, gives an MDS upper bound, and covers parameter pockets not certified by the specialized BCHKS estimate used below. We make the literature implication integral, transport it exactly to standard circle codes admitting the displayed stereographic or diagonal presentation, and record positive-density families without promoting any of these statements to a near-capacity theorem. The unrestricted smooth/circle frontier beyond half distance still requires new information about lists or received-line incidence.

Theorem 1.1 (Self-contained quadratic target theorem). *Let $C_n = \text{RS}_{\mathbb{F}_n}(D_n, k_n)$ be any Reed-Solomon row sequence over arbitrary characteristics, let $\emptyset \neq \Gamma_n \subseteq \mathbb{F}_n$, and put $B_n^* = \lfloor \varepsilon_n |\Gamma_n| \rfloor$ as in (1.2).*

- (i) *If $B_n^* = 0$, no agreement is safe and the set defining a_n^* is empty.*
- (ii) *If $B_n^* = |\Gamma_n|$, then $a_n^* = k_n + 1$.*
- (iii) *Suppose eventually*

$$(1.3) \quad 1 \leq B_n^* < |\Gamma_n|, \quad B_n^* \leq n - k_n - 1, \quad (B_n^* - 1)^2 \geq n(3(B_n^* - 1) - (n - k_n)).$$

Then the threshold exists and is exactly

$$(1.4) \quad a_n^* = n - B_n^* + 1, \quad \delta_n^* = \frac{B_n^* - 1}{n}.$$

In particular, if $k_n/n \rightarrow \rho$,

$$\frac{B_n^*}{n} \rightarrow b < \alpha_\rho := \frac{3 - \sqrt{5 + 4\rho}}{2},$$

and $1 \leq B_n^ < |\Gamma_n|$ eventually, then $\delta_n^* \rightarrow b$. This conclusion applies, without a smoothness, odd-characteristic, Fourier, atlas, or ray-compiler hypothesis, to every multiplicative smooth RS row and to every circle code that has been proved diagonally agreement-equivalent to an RS presentation.*

Proof. If $B_n^* = 0$, the universal tangent floor at agreement n is one, so no grid point is safe. If $B_n^* = |\Gamma_n|$, the numerator is always at most the size of the challenge set, so the first grid point $k_n + 1$ is safe. In the remaining case put $a_0 = n - B_n^* + 1$. Then $n - a_0 = B_n^* - 1$, and the last inequality in (1.3) places a_0 in the mean-overlap range. The exact numerator of Corollary 4.7 is $\min\{|\Gamma_n|, B_n^*\} = B_n^*$, so a_0 is safe. The second inequality in (1.3) puts $a_0 - 1$ on the admissible grid, and Proposition 4.2 gives at least $\min\{|\Gamma_n|, B_n^* + 1\} = B_n^* + 1$ bad slopes there. Hence $a_0 - 1$ is unsafe, and monotonicity proves (1.4). The asymptotic specialization follows because, after division by n^2 , its strict limiting margin is $b^2 - 3b + 1 - \rho > 0$. Moreover $\alpha_\rho < 1 - \rho$ for $0 < \rho < 1$, since this is equivalent to $\sqrt{5 + 4\rho} > 1 + 2\rho$. Thus $B_n^* \leq n - k_n - 1$ eventually, and hence (1.3) holds for all sufficiently large n . $\square$

Theorem 1.2 (Positive-density exact MCA staircase). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, put $R = n - k$, and let $\emptyset \neq \Gamma \subseteq \mathbb{F}$.*

- (i) *If $r \in \mathbb{Z}$ and*

$$(1.5) \quad 3r \geq R + 1, \quad 2r < R, \quad n(R - r) < r(r + 2)(R - 2r),$$

then

$$(1.6) \quad B_{C, \Gamma}^{\text{MCA}}(n - r) = \min\{|\Gamma|, r + 1\}.$$

Together with the quadratic theorem, this gives an exact finite staircase from radius zero through the union of the two displayed ranges.

- (ii) *If $k_n/n \rightarrow \rho \in (0, 1)$ and the integers r_n satisfy $r_n/n \rightarrow \tau$ with $0 < \tau < (1 - \rho)/2$, then*

$$B_{C_n, \Gamma_n}^{\text{MCA}}(n - r_n) = \min\{|\Gamma_n|, r_n + 1\}$$

for every RS row sequence and every nonempty challenge sequence, once n is sufficiently large. Hence exact radii have normalized lower density at least $(1 - \rho)/2$ among all Hamming radii and relative density one inside the unique-decoding interval.

- (iii) *For every fixed dyadic n , the multiplicative smooth conclusion holds over a set of primes of relative prime density $1/\varphi(n)$. Standard bivariate circle codes of odd dimension over $\mathbb{F}_{p^2}$, indexed by base primes p , occur over a prime set of density $1/\varphi(2n)$, and Chebyshev-smooth circle x -domain RS rows occur over a prime set of density $1/\varphi(4n)$. These are fixed-length densities, not a positive-density growing prime-field family.*

- (iv) At $n = 32, k = 15, r = 6$, the quadratic inequality is strict enough to give $B_{C,\Gamma}^{\text{MCA}}(26) = \min\{|\Gamma|, 7\}$, although $3r > R$ and the specialized estimate (LC5) does not certify this value. In particular it gives smooth subgroup rows for every $p \equiv 1 \pmod{32}$, a prime set of density $1/16$, and standard circle cosets for every $p \equiv -1 \pmod{64}$, a prime set of density $1/32$.

Proof. Part (i) is Theorem 5.2; part (ii) is Corollary 5.4; and parts (iii)–(iv) are Theorem 8.23 and Corollary 8.24. $\square$

Remark 1.3 (Meaning and limitation of density). The word “density” in Theorem 1.2 is used in two explicit senses: normalized density of radii in part (ii) and relative density inside the primes in part (iii). It does not assert that a positive proportion of slopes is bad near capacity. Neither the presently available profile-envelope estimates nor the literature currently supplies such an all-received-line near-capacity theorem. Moreover the fixed-length prime densities tend to zero with n ; Proposition 8.27 proves that this loss cannot be removed for growing dyadic prime-field domains.

Theorem 1.4 (Certified Proximity-Prize rows at all four rates). *Fix the affine MCA sampler of [ABF26], in which γ is uniform in $\mathbb{F}$, and the target $\varepsilon^* = 2^{-128}$. For each $\rho \in \{1/2, 1/4, 1/8, 1/16\}$ there is an explicitly certified prime $p < 2^{256}$, a power of two n , and the multiplicative subgroup $D \leq \mathbb{F}_p^*$ of order n , such that $k = \rho n = 2^{40}$ and the complete real-radius safe set of $C = \text{RS}_{\mathbb{F}_p}(D, k)$ is*

$$(P1) \quad \{\delta \in [0, 1] : e_{\text{MCA}}(C, \delta) \leq 2^{-128}\} = \left[0, \frac{B}{n}\right), \quad B = \lfloor p 2^{-128} \rfloor.$$

The endpoint B/n is unsafe. Equivalently, the largest safe integer radius is $B - 1$, the first unsafe integer radius is B , and

$$(P2) \quad B_C^{\text{MCA}}(n - B + 1) = B, \quad B_C^{\text{MCA}}(n - B) \geq B + 1.$$

The four transition numerators are

	ρ	n	B	$\delta_{C,\text{sup}}^*(2^{-128})$
(P3)	1/2	2^{41}	389500552609	$389500552609/2^{41}$
	1/4	2^{42}	1210584858040	$1210584858040/2^{42}$
	1/8	2^{43}	2879806199253	$2879806199253/2^{43}$
	1/16	2^{44}	6233898019554	$6233898019554/2^{44}$

These are smooth-domain rows in the literal sense of the challenge: the evaluation group has power-of-two order, the dimension is at the stated upper limit, and the field has fewer than 2^{256} elements. The numerical consequences are also implied by [BCHKS25, Cor. 1.4] together with the below-half decomposition; their role here is to provide independent exact certificates and target bookkeeping, not a priority claim.

Proof. The mean-overlap staircase Corollary 4.11 gives the exact safe numerator $B = r_\rho(n) + 1$ at radius $B - 1$, while the universal tangent floor gives at least $B + 1$ slopes at radius B . The explicit primes, their Proth certificates, and the exact budget divisions are given and checked in Table 1 and Proposition 6.1. Monotonicity and $r(\delta) = \lfloor \delta n \rfloor$ then give (P1). The displayed fractions are the four values of B/n . $\square$

Corollary 1.5 (An extension-field smooth row). *Let $D \leq \mathbb{F}_{17^{32}}^*$ have order 512 and let $C = \text{RS}_{\mathbb{F}_{17^{32}}}(D, 256)$. At target 2^{-128} ,*

$$\{\delta \in [0, 1] : e_{\text{MCA}}(C, \delta) \leq 2^{-128}\} = [0, 3/256),$$

the largest safe integer radius is 5, and radius 6 is the first unsafe one. In particular the supremal transition is $3/256$, and it is not itself safe.

Proof. Here $17^{32} = 2367911594760467245844106297320951247361$ and $\lfloor 17^{32} 2^{-128} \rfloor = 6$. Since $3(6 - 1) \leq 512 - 256$, Corollary 4.3 gives numerator 6 at radius 5, while Proposition 4.2 gives

at least 7 bad slopes at radius 6. The endpoint statement follows from the closed integer-ball convention. $\square$

Theorem 1.6 (Certificate threshold formula). *Let $C_n = \text{RS}_{\mathbb{F}_n}(D_n, k_n)$, let $\emptyset \neq \Gamma_n \subseteq \mathbb{F}_n$, and suppose $k_n/n \rightarrow \rho$. Let $B_n^* = \lfloor \varepsilon_n |\Gamma_n| \rfloor$, and assume the first safe agreement a_n^* in (1.2) exists. Suppose there is a sequence $g_n \in [0, 1 - k_n/n]$ such that, for every fixed $\eta > 0$, all admissible integer agreements*

$$a \geq k_n + 1 + (g_n + \eta)n$$

are eventually safe and all admissible integer agreements

$$a \leq k_n + 1 + (g_n - \eta)n$$

are eventually unsafe. Then

$$(1.7) \quad a_n^* = k_n + 1 + g_n n + o(n), \quad \delta_n^* = 1 - \rho - g_n + o(1).$$

Assume in addition that $D_n \subseteq \mathbb{B}_n \subseteq \mathbb{F}_n$ with $\mathbb{B}_n$ a finite subfield and $\log |\mathbb{B}_n| = o(n)$, and put $\beta_n = \log_2 |\mathbb{B}_n|$ and $\tau_n = n^{-1} \log_2(1 + B_n^)$, and define the target-adjusted identity crossing*

$$(1.8) \quad g_{T,n} = \sup \left(\{g \in [0, 1 - k_n/n] : H_2(k_n/n + g) - \beta_n g \geq \tau_n\} \cup \{0\} \right).$$

If matching upper and lower certificates satisfy the preceding safe and unsafe conditions with $g_n = g_{T,n} + o(1)$, then

$$(1.9) \quad \delta_n^* = 1 - \rho - g_{T,n} + o(1).$$

When $\tau_n = o(1)$ and $\rho_n = k_n/n \rightarrow \rho \in (0, 1)$, put

$$(1.10) \quad g_{0,n} = \sup \{g \in [0, 1 - \rho_n] : H_2(\rho_n + g) \geq \beta_n g\}.$$

Then $g_{T,n} = g_{0,n} + o(1)$, so $\delta_n^ = 1 - \rho - g_{0,n} + o(1)$.*

Proof. For every fixed $\eta > 0$, monotonicity places the first safe agreement between

$$k_n + 1 + (g_n - \eta)n + O(1) \quad \text{and} \quad k_n + 1 + (g_n + \eta)n + O(1).$$

Divide by n , let $n \rightarrow \infty$, and then let $\eta \downarrow 0$. Since $\delta_n^* = 1 - a_n^*/n$, this proves (1.7). The second assertion is the same argument with the displayed crossing substituted for g_n .

For completeness, the identity-prefix mean at $a = k_n + 1 + gn + O(1)$ is

$$\binom{n}{a} |\mathbb{B}_n|^{-(a-k_n-1)}.$$

Stirling's formula gives its normalized logarithm as

$$\frac{1}{n} \log_2 \left(\binom{n}{a} |\mathbb{B}_n|^{-(a-k_n-1)} \right) = H_2(\rho_n + g) - \beta_n g + o(1),$$

uniformly on compact subintervals of $0 < \rho_n + g < 1$. This explains (1.8); it does not supply the assumed upper certificate. Finally, $f_n(g) = H_2(\rho_n + g) - \beta_n g$ is strictly concave, $f_n(0) = H_2(\rho_n) > 0$, and $f_n(1 - \rho_n) < 0$. Eventually $\tau_n < f_n(0)$. Hence the τ_n -superlevel set is an interval containing zero, with right endpoint $g_{T,n}$, whereas the zero-superlevel set has right endpoint $g_{0,n}$; in particular, $0 \leq g_{0,n} - g_{T,n}$. Fix $\eta > 0$. If $g_{0,n} < \eta$, this difference is already less than η . Otherwise, if $g_{0,n} - g_{T,n} \geq \eta$, concavity on $[0, g_{0,n}]$ gives

$$f_n(g_{0,n} - \eta) \geq \frac{\eta}{g_{0,n}} f_n(0) \geq \eta H_2(\rho_n).$$

The left side is at most τ_n , by the definition of the right τ_n -crossing, while the right side is bounded away from zero. This contradicts $\tau_n = o(1)$. Hence $g_{T,n} = g_{0,n} + o(1)$. $\square$

Remark 1.7 (Scope of the entropy crossing). The first part of Theorem 1.6 is unconditional once matching certificates have been proved. The identity crossing in (1.9) is a specialization, not an identity-only theorem for every smooth or circle domain. Quotient, subfield, planted, and determinant-ray profiles can have a larger realized-image scale; in such a row g_n is the crossing of the full certified profile envelope. Support or support-pair estimates also require a separate

projection to distinct slopes. These are precisely the two points at which an unrestricted near-capacity claim would need additional input.

2. MCA WITNESSES AND EXACT SUPPORT REDUCTION

Throughout this section $\mathbb{F}$ is a finite field, $D \subseteq \mathbb{F}$ has size n , and $C = \text{RS}_{\mathbb{F}}(D, k)$. Write $\mathbb{F}[X]_{<k} = \{f \in \mathbb{F}[X] : \deg f < k\}$, and let $\text{ev}_S : \mathbb{F}[X]_{<k} \rightarrow \mathbb{F}^S$ be restriction of the evaluation map.

Definition 2.1 (Explanation and support-wise nontriviality). *For $u \in \mathbb{F}^D$ and $S \subseteq D$, an explanation of u on S is a polynomial $f \in \mathbb{F}[X]_{<k}$ with $\text{ev}_S(f) = u|_S$. We write $\text{Expl}_C(u; S)$ when such an f exists. A pair $r = (r_0, r_1)$ is simultaneously explained on S when $\text{Expl}_C(r_0; S)$ and $\text{Expl}_C(r_1; S)$ both hold, with no requirement that the two explaining polynomials coincide. It is support-wise nontrivial on S when it is not simultaneously explained there. We denote this last condition by $\text{NT}_C(r; S)$.*

Thus support-wise nontriviality is a condition on the same support that explains the affine combination. It does not mean that both coordinates are individually unexplained; failure of either explanation is enough.

Definition 2.2 (MCA-bad slope). *Let $r = (r_0, r_1) \in (\mathbb{F}^D)^2$ and let a be an agreement threshold. A finite slope $\gamma \in \mathbb{F}$ is support-wise MCA-bad at agreement a if there are $S \subseteq D$, $|S| \geq a$, and $h \in \mathbb{F}[X]_{<k}$ such that $\text{ev}_S(h) = (r_0 + \gamma r_1)|_S$ and $\text{NT}_C(r; S)$ holds. Define*

$$B_C^{\text{MCA}}(a) = \max_{r \in (\mathbb{F}^D)^2} |\{\gamma \in \mathbb{F} : \gamma \text{ is MCA-bad for } r \text{ at } a\}|.$$

The official MCA maximum mixes two logically different sources: received pairs that are globally far from every codeword pair, and pairs that are close to one codeword pair but fail on a slope-dependent support. The following definitions and theorem separate them exactly.

Definition 2.3 (CA numerator and sparse mutual numerator). *Assume $0 \leq a \leq n$ and put $t = n - a$. A received pair (r_0, r_1) is column-far at agreement a if it is not simultaneously explained on any support of size at least a . Let $B_C^{\text{CA}}(a)$ be the maximum, over column-far pairs, of the number of slopes γ for which $r_0 + \gamma r_1$ is explained on some support of size at least a . If there is no column-far pair, this maximum is defined to be zero. Define the sparse mutual numerator*

$$(\text{SP1}) \quad \sigma_C(a) = \max \{ \#\{\gamma : \gamma \text{ is MCA-bad for } (e_0, e_1)\} : |\text{supp}(e_0) \cup \text{supp}(e_1)| \leq t \}.$$

Thus $B_C^{\text{CA}}(a)/|\mathbb{F}|$ is the ordinary affine-line correlated agreement error on the integer grid, while $\sigma_C(a)$ records the purely mutual obstruction after translation by a nearby codeword pair.

Theorem 2.4 (Exact sparsification of the mutual layer). *For every linear code $C \subseteq \mathbb{F}^D$ and every agreement threshold $0 \leq a \leq n$,*

$$(\text{SP2}) \quad B_C^{\text{MCA}}(a) = \max\{B_C^{\text{CA}}(a), \sigma_C(a)\}.$$

Equivalently, after division by $|\mathbb{F}|$, the official MCA error is the maximum of the CA error and the sparse mutual error.

Proof. Both quantities on the right are obtained by maximizing the MCA-bad set over restricted classes of pairs, so the left side is at least their maximum. Conversely fix a received pair. If it is column-far, every support that explains a line point necessarily fails to explain the pair; its MCA-bad set is therefore its CA-bad set and has size at most $B_C^{\text{CA}}(a)$.

Otherwise choose codewords $c_0, c_1 \in C$ and a common support of size at least a , and put $e_i = r_i - c_i$. The union of the supports of e_0, e_1 has size at most $n - a = t$. At each slope, translation sends an explaining codeword h to $h - c_0 - \gamma c_1$, and is a bijection between witness supports for (r_0, r_1) and for (e_0, e_1) . It also preserves simultaneous explanation on that support. Hence the two pairs have exactly the same MCA-bad slopes, and their number is at most $\sigma_C(a)$. $\square$

For a nonempty challenge set $\Gamma \subseteq \mathbb{F}$, write $B_{C,\Gamma}^{\text{CA}}(a)$ and $\sigma_{C,\Gamma}(a)$ for the same maxima with the bad slopes restricted to Γ . The proof just given is unchanged after intersecting every bad-slope set with Γ , and therefore gives the exact challenge-restricted identity

$$(SP3) \quad B_{C,\Gamma}^{\text{MCA}}(a) = \max\{B_{C,\Gamma}^{\text{CA}}(a), \sigma_{C,\Gamma}(a)\}.$$

Theorem 2.5 (Exact mutual layer below half the distance). *Let $C \subseteq \mathbb{F}^D$ be a linear code of minimum distance $d_{\min}$, let $\emptyset \neq \Gamma \subseteq \mathbb{F}$, and let $0 \leq r \leq n$ satisfy $2r \leq d_{\min} - 1$. Then*

$$(HD1) \quad \sigma_{C,\Gamma}(n-r) = \min\{|\Gamma|, r\}, \quad B_{C,\Gamma}^{\text{MCA}}(n-r) = \max\{B_{C,\Gamma}^{\text{CA}}(n-r), \min\{|\Gamma|, r\}\}.$$

For a Reed–Solomon row the hypothesis is $2r \leq n - k$.

Proof. Fix a sparse pair (e_0, e_1) whose support union E has size at most r , and let γ be bad on a support S of size at least $n - r$, explained there by a codeword $c \in C$. On $S \setminus E$ the sparse line point is zero, so $\text{supp}(c) \subseteq E \cup (D \setminus S)$ has size at most $2r < d_{\min}$. Thus $c = 0$. If $S \cap E$ contained no nonzero pair coordinate, the zero codeword pair would simultaneously explain (e_0, e_1) on S , contrary to badness. Hence some $x \in S \cap E$ satisfies

$$(e_0(x), e_1(x)) \neq (0, 0), \quad e_0(x) + \gamma e_1(x) = 0.$$

Necessarily $e_1(x) \neq 0$, and this coordinate determines the single slope $-e_0(x)/e_1(x)$. Therefore every sparse pair has at most $\min\{|\Gamma|, |E|\} \leq \min\{|\Gamma|, r\}$ bad slopes.

For the reverse bound put $t = \min\{|\Gamma|, r\}$, choose distinct $x_1, \dots, x_t \in D$ and distinct $\gamma_1, \dots, \gamma_t \in \Gamma$, and set $e_1(x_i) = 1$, $e_0(x_i) = -\gamma_i$, with both words zero elsewhere. For $S_i = D \setminus \{x_j : j \neq i\}$, the line point of slope γ_i is zero on S_i , and $|S_i| = n - t + 1 \geq n - r$. The restriction of e_1 to S_i cannot be explained by a codeword: such a codeword would be nonzero at x_i and zero on $n - t$ coordinates, hence would have weight at most $t < d_{\min}$. Thus every γ_i is MCA-bad and equality holds. The second formula is (SP3); an RS code has $d_{\min} = n - k + 1$. $\square$

Lemma 2.6 (Pair-interleaved Johnson bound). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, let $1 \leq A \leq n$, and suppose $A^2 > (k - 1)n$. For every received pair $y \in (\mathbb{F}^D)^2$, the number of codeword pairs $(p_0, p_1) \in C^2$ simultaneously agreeing with y on at least A coordinates is at most*

$$(HD2) \quad J_2(A) = \left\lfloor \frac{n(A - k + 1)}{A^2 - (k - 1)n} \right\rfloor.$$

Proof. Let the list have size L , and for every member retain exactly A agreement coordinates. If d_x is the number of retained sets containing x , then $\sum_x d_x = LA$. Two distinct codeword pairs can agree coordinatewise on at most $k - 1$ points, since at least one nonzero difference polynomial has degree less than k . Therefore

$$\frac{L^2 A^2 / n - LA}{2} \leq \sum_{x \in D} \binom{d_x}{2} \leq \binom{L}{2} (k - 1),$$

where the first inequality is Cauchy–Schwarz. Rearrangement gives $L(A^2 - (k - 1)n) \leq n(A - k + 1)$, and integrality gives (HD2). $\square$

Theorem 2.7 (Self-contained half-Johnson CA and MCA bound). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, let $\emptyset \neq \Gamma \subseteq \mathbb{F}$, and let r be an integer with $0 \leq r \leq n$. Put $A = n - 2r$. If $A > 0$ and $A^2 > (k - 1)n$, then*

$$(HD3) \quad B_{C,\Gamma}^{\text{CA}}(n-r) \leq \min\{|\Gamma|, 1 + (r + 1)J_2(A)\}.$$

If in addition $2r \leq n - k$, the same upper bound holds for $B_{C,\Gamma}^{\text{MCA}}(n-r)$.

Proof. Fix a column-far received pair and let $Z \subseteq \Gamma$ be its CA-bad set. The assertion is immediate when $|Z| \leq 1$, so fix an anchor $\gamma_0 \in Z$. For every $\gamma \in Z \setminus \{\gamma_0\}$, choose explaining codewords c_{γ_0}, c_{γ} and agreement sets of size at least $n - r$, and define

$$P_1 = \frac{c_{\gamma_0} - c_{\gamma}}{\gamma_0 - \gamma}, \quad P_0 = c_{\gamma_0} - \gamma_0 P_1.$$

On the intersection of the two agreement sets, of size at least $n - 2r = A$, the codeword pair (P_0, P_1) agrees with the received pair. Hence all such pairs lie in a list of size at most $J_2(A)$ by Lemma 2.6.

Fix one pair in that list and let E be its column-error set, of size e . Column-farness gives $e \geq r + 1$. If a slope γ rides this pair, meaning its explaining codeword is $P_0 + \gamma P_1$, then the nonzero affine coordinate errors vanish at at least $e - r$ coordinates of E . Each coordinate vanishes for at most one finite slope, so double counting gives at most $e/(e - r) \leq r + 1$ riders. Summing over the pair list and adding the anchor proves (HD3). Under $2r \leq n - k$, Theorem 2.5 says that the other branch of MCA has only $\min\{|\Gamma|, r\}$ slopes. Since $n(A - k + 1) - (A^2 - (k - 1)n) = A(n - A) \geq 0$, one has $J_2(A) \geq 1$, so this tangent term is no larger than the displayed CA bound. $\square$

Remark 2.8 (The exact remaining prize certificate). At a proposed safe agreement a_+ , an unrestricted finite-row proof must establish both

$$B_C^{\text{CA}}(a_+) \leq B^*, \quad \sigma_C(a_+) \leq B^*, \quad B^* = \left\lfloor 2^{-128} |\mathbb{F}| \right\rfloor.$$

An unsafe construction at $a_+ - 1$ does not prove either upper bound. This identity is the shortest exact description of what remains for the near-capacity rows not covered by Corollary 5.6; the older quadratic compiler Corollary 4.12 is a self-contained subrange.

The finite-slope restriction costs at most the point at infinity, which may also be treated by exchanging the two coordinates. The support-wise condition is essential: explanation and nontriviality are tested on the same set.

Definition 2.9 (Exact-agreement witness incidence). *Assume $k + 1 \leq a \leq n$. For a received line $r = (r_0, r_1)$, its exact- a witness incidence is*

$$(2.1) \quad \mathcal{W}_a(r) = \left\{ (\gamma, S, h) : \begin{array}{l} \gamma \in \mathbb{F}, \quad S \in \binom{D}{a}, \quad h \in \mathbb{F}[X]_{<k}, \\ \text{ev}_S(h) = (r_0 + \gamma r_1)|_S, \quad \text{NT}_C(r; S) \end{array} \right\}.$$

Its slope projection is $\pi_\gamma(\gamma, S, h) = \gamma$, and its exact bad-slope set is $Z_a(r) = \pi_\gamma(\mathcal{W}_a(r)) \subseteq \mathbb{F}$.

Lemma 2.10 (Reduction to exact agreement). *For $a \geq k + 1$, the set $Z_a(r)$ is precisely the set of slopes that are MCA-bad at threshold a in Definition 2.2.*

Proof. An exact- a witness is a threshold- a witness. Conversely, let (γ, S, h) witness badness with $|S| \geq a$. At least one of $r_0|_S, r_1|_S$ is outside $\text{ev}_S(\mathbb{F}[X]_{<k})$; call it $u|_S$. Interpolate u on any fixed k -subset $K \subseteq S$. If the resulting polynomial agreed with u at every point of $S \setminus K$, it would explain u on S , a contradiction. Hence some $x \in S \setminus K$ makes u unexplained on $K \cup \{x\}$. Extend this set to an a -subset $S' \subseteq S$. Non-explanation persists on supersets, while the same h explains $r_0 + \gamma r_1$ on S' . $\square$

Proposition 2.11 (Exact support-atlas upper bound). *For every Reed–Solomon row, every nonempty challenge set $\Gamma \subseteq \mathbb{F}$, and every $k + 1 \leq a \leq n$,*

$$(2.2) \quad B_{C, \Gamma}^{\text{MCA}}(a) \leq \min \left\{ |\Gamma|, \binom{n}{a} \right\}.$$

This is a literal finite-row bound: it has no polynomial or subexponential loss.

Proof. By Lemma 2.10, choose one noncommon exact- a support for every bad slope. One noncommon support carries at most one slope, since two distinct explained slopes would separately explain the two received words on that support. The chosen-support map is therefore injective. Intersect with the challenge set. $\square$

3. EXACT SYNDROME GEOMETRY AND THE DEEP REGIME

The *syndrome* of a word is the result of applying a parity-check matrix; it vanishes exactly on codewords. Thus syndrome space forgets the part of a received word already lying in the code and retains only its error class. This description separates possible error supports from

distinct line parameters. Put $R = n - k$, the redundancy (and syndrome-space dimension), and choose the generalized Vandermonde parity-check matrix H with columns

$$(3.1) \quad h_x = \lambda_x(1, x, \dots, x^{R-1})^\top, \quad \lambda_x = \left(\prod_{y \in D \setminus \{x\}} (x - y) \right)^{-1}.$$

Lagrange interpolation gives $\sum_{x \in D} \lambda_x x^j = 0$ for $0 \leq j \leq n - 2$, so this matrix annihilates every degree-less-than- k evaluation. Every set of at most R columns is independent. For $E \subseteq D$, write $V_E = \text{Span}\{h_x : x \in E\}$; it is exactly the set of syndromes of words supported on E . We write $s(u) = Hu^\top$ for the syndrome of u .

Proposition 3.1 (Syndrome-line normal form). *Let $S = D \setminus E$. A word u is explained on S if and only if $s(u) \in V_E$. Consequently γ is MCA-bad on S for a pair (u_0, u_1) if and only if*

$$(3.2) \quad s(u_0) + \gamma s(u_1) \in V_E, \quad \{s(u_0), s(u_1)\} \not\subseteq V_E.$$

For fixed E , there is at most one such finite slope.

Proof. If $u = c + e$, where $c \in C$ and e is supported on E , then $s(u) = s(e) \in V_E$. Conversely, if $s(u) = \sum_{x \in E} e_x h_x$, subtract the word supported on E with values e_x ; the result has zero syndrome and belongs to C . Apply this equivalence to the line point and to both coordinates. Passing (3.2) to $\mathbb{F}^R/V_E$ shows that two distinct solutions would put both projected syndromes at zero, contradicting badness. $\square$

Since $\lambda_x \neq 0$, one has $[h_x] = [1 : x : \dots : x^{R-1}]$; these are the points of the rational normal curve in projective syndrome space, while the weights λ_x merely choose vector representatives. A *secant span* is the linear span V_E of a selected set of those points. We call the meeting of a syndrome line with V_E *transverse* when the entire line is not contained in V_E , equivalently when its two generators are not both in V_E . The next result is a *compiler* in the sense that it translates the bad-slope problem, exactly and in both directions, into this line-secant incidence problem. A *ray* means a one-dimensional direction in syndrome or coefficient space; different raw witnesses may represent the same ray and hence the same slope.

Theorem 3.2 (Exact syndrome-secant compiler). *Put $t = n - a$, and for a syndrome line $\ell_{y_0, y_1} = \{y_0 + \gamma y_1 : \gamma \in \mathbb{F}\}$ define*

$$(3.3) \quad \Theta_t(y_0, y_1) = |\{\gamma \in \mathbb{F} : \exists E \subseteq D, |E| \leq t, y_0 + \gamma y_1 \in V_E, \{y_0, y_1\} \not\subseteq V_E\}|.$$

Then

$$(3.4) \quad B_C^{\text{MCA}}(a) = \max_{y_0, y_1 \in \mathbb{F}^{n-k}} \Theta_t(y_0, y_1).$$

For each fixed E , the affine line has at most one transverse intersection parameter with V_E . For a Reed-Solomon code the spaces V_E are the spans of at most t points of the weighted rational normal curve in (3.1). Thus the higher-dimensional ray problem is exactly a transverse line-secant incidence problem, with repeated intersection parameters deduplicated.

Proof. Apply Proposition 3.1 with $S = D \setminus E$, and take the union over $|E| \leq t$. The syndrome map is surjective, so maximizing over received pairs is the same as maximizing over (y_0, y_1) . In the quotient $\mathbb{F}^{n-k}/V_E$, a transverse intersection solves $\bar{y}_0 + \gamma \bar{y}_1 = 0$ with the two projected vectors not both zero, and therefore has at most one finite solution. The last assertion is the Vandermonde description of the parity columns. $\square$

A *parity-column circuit* is a minimally linearly dependent set of columns of H . The next lemma says that many distinct transverse rays cannot all be supported inside an independent union of columns.

Lemma 3.3 (Independent-union ray bound). *Let every set of at most R parity columns be independent, put $t < R$, and fix a syndrome line. For each slope in a transverse set Z , choose an error set E_γ , $|E_\gamma| \leq t$, witnessing (3.3). If $U = \bigcup_{\gamma \in Z} E_\gamma$ has size at most R , then $|Z| \leq t + 1$. Consequently, more than $t + 1$ transverse slopes force the witness union to contain a parity-column circuit.*

Proof. If $|Z| \leq 1$, there is nothing to prove; assume that Z contains two distinct slopes. Two distinct line points lie in V_U , hence $y_0, y_1 \in V_U$. Column independence gives unique coefficient lifts $e_0, e_1 \in \mathbb{F}^U$, and every chosen lift is $e_\gamma = e_0 + \gamma e_1$. Let U' be the set of s coordinates where the pair (e_0, e_1) is nonzero. If $s \leq t$ and E_γ contained all of U' , then $y_0, y_1 \in V_{U'} \subseteq V_{E_\gamma}$, contradicting the transversality of this particular witness. Hence some $x \in U' \setminus E_\gamma$ exists, and uniqueness of the lift gives $e_0(x) + \gamma e_1(x) = 0$. Each nonzero affine coordinate function has at most one root, so $|Z| \leq s \leq t$. If $s > t$, every chosen lift of weight at most t has at least $s - t$ zero coordinates. Double counting the pairs (γ, x) with $e_0(x) + \gamma e_1(x) = 0$ gives $|Z|(s - t) \leq s$, whence $|Z| \leq \lfloor s/(s - t) \rfloor \leq t + 1$. If $|U| \leq R$, independence holds; the contrapositive proves the circuit assertion. $\square$

Theorem 3.4 (Bounded residual-kernel ray compiler). *Let H be a parity-check matrix over $\mathbb{F}$, let U be a set of columns, and put*

$$\kappa(U) = \dim_{\mathbb{F}} \ker(H_U : \mathbb{F}^U \longrightarrow \mathbb{F}^{n-k}).$$

Fix a syndrome line $y_0 + \gamma y_1$, an integer $t \geq 0$, and a set Z of finite transverse slopes. Suppose that for every $\gamma \in Z$ there is $E_\gamma \subseteq U$, $|E_\gamma| \leq t$, such that

$$y_0 + \gamma y_1 \in V_{E_\gamma}, \quad \{y_0, y_1\} \not\subseteq V_{E_\gamma}.$$

Then

$$(RC_{\ker}) \quad |Z| \leq (t + 1)|\mathbb{F}|^{\kappa(U)}.$$

For a Reed–Solomon parity check of redundancy $R = n - k$, $\kappa(U) = \max\{0, |U| - R\}$. Hence a certified residual chart with $(|U| - R)_+ \log|\mathbb{F}| = o(n)$ has a direct subexponential ray payment with the displayed exact finite loss.

Proof. The assertion is immediate when $|Z| \leq 1$. Otherwise two line points belong to V_U , so $y_0, y_1 \in V_U$. Choose lifts $b_0, b_1 \in \mathbb{F}^U$ with $H_U b_i = y_i$. For every γ , choose a lift c_γ supported in E_γ and put $z_\gamma = c_\gamma - b_0 - \gamma b_1 \in \ker H_U$.

Fix $z \in \ker H_U$, and consider the slopes for which $z_\gamma = z$. Put

$$U_z = \{x \in U : (b_0(x) + z(x), b_1(x)) \neq (0, 0)\}, \quad s = |U_z|.$$

Every active coordinate is a nonzero affine function of γ , and therefore vanishes for at most one slope. Hence the total number of active zero incidences in this kernel class is at most s . If $s > t$, the weight bound on c_γ forces at least $s - t$ active zeros, so

$$|Z_z|(s - t) \leq s, \quad |Z_z| \leq \lfloor s/(s - t) \rfloor \leq t + 1.$$

If $s \leq t$, transversality forces at least one active zero: without one, $U_z = \text{supp } c_\gamma \subseteq E_\gamma$, while both $b_0 + z$ and b_1 are supported in U_z , putting y_0, y_1 in V_{E_γ} . Thus this kernel class contains at most $s \leq t$ slopes. Summing over the $|\mathbb{F}|^{\kappa(U)}$ kernel elements proves the bound. The MDS column property gives the Reed–Solomon formula for $\kappa(U)$. $\square$

Theorem 3.5 (Single MDS-circuit ray compiler). *Let H be a Reed–Solomon parity check of redundancy R , let $U \subseteq D$ have size $R + 1$, put $t < R$, and fix a syndrome line $y_0 + \gamma y_1$. If every slope in a transverse set Z has a witnessing error set $E_\gamma \subseteq U$ of size at most t , then*

$$(RC_{\text{circ}}) \quad |Z| \leq \binom{R + 1}{2}.$$

Thus one fixed MDS-circuit profile has a field-independent polynomial ray payment.

Proof. There is nothing to prove when $|Z| \leq 1$. Otherwise two distinct line points lie in V_U ; subtracting them and solving for the constant term gives $y_0, y_1 \in V_U$. Choose lifts $b_0, b_1 \in \mathbb{F}^U$. The circuit kernel is generated by a vector z whose coordinates are all nonzero, since a zero coordinate would give a proper dependent subset of the minimal circuit U . For each slope choose $c_\gamma \in \mathbb{F}$ so that $e_\gamma = b_0 + \gamma b_1 + c_\gamma z$ is supported in E_γ . For $x \in U$, set

$$f_x(\gamma) = -(b_0(x) + \gamma b_1(x))/z(x).$$

A zero coordinate is an equality $c_\gamma = f_x(\gamma)$, and at least $R + 1 - t \geq 2$ such equalities hold.

Every transverse slope makes two nonidentical functions $f_x, f_{x'}$ equal. Indeed, otherwise all vanishing coordinates belong to one class C of identical affine functions, say $f_x = \alpha + \beta\gamma$ on C , and no function outside C has that value at γ . Then $\text{supp } e_\gamma = U \setminus C \subseteq E_\gamma$, while $b_0 + \alpha z$ and $b_1 + \beta z$ are also supported in $U \setminus C$. This puts both y_0, y_1 in V_{E_γ} , a contradiction. Two nonidentical affine functions agree at at most one finite slope. Charging each slope to one unordered pair proves the displayed bound. $\square$

4. THE MEAN-OVERLAP REGIME

For a fixed received line, the exact bad-slope set is therefore the set of unique cancelling line parameters contributed by the transverse quotient equations. For each E , $|E| \leq n - a$, project the two syndromes to $\mathbb{F}^R/V_E$. If the second projection is nonzero and the first is a scalar multiple of it, record the unique cancelling scalar; otherwise record nothing. Deduplicating these scalars gives the numerator. In particular, the number of supports, the number of pairs of supports, and the number of distinct slopes are three different quantities.

We call $3(n - a) \leq d - 1$ the *deep regime*: the allowed error radius $n - a$ is so small relative to the minimum codeword weight d that the union of three permitted error supports still cannot contain a nonzero codeword support.

Theorem 4.1 (Deep-regime upper bound). *Let $C \subseteq \mathbb{F}^n$ be linear with minimum nonzero weight d , let $0 \leq a \leq n$, put $r = n - a$, and assume $3r \leq d - 1$. Then*

$$(3.5) \quad B_C^{\text{MCA}}(a) \leq r + 1.$$

Proof. Fix (u_0, u_1) , and let Z be the slopes for which $u_0 + \gamma u_1$ is within distance r of C . If $|Z| \leq r + 1$, there is nothing to prove. Otherwise choose distinct $\gamma_1, \gamma_2 \in Z$, codewords p_1, p_2 , and error sets E_1, E_2 , each of size at most r , such that $u_0 + \gamma_i u_1 = p_i$ off E_i . Solving the two affine equations gives $c_0, c_1 \in C$ for which the error pair $e_i = u_i - c_i$ is supported on $T = E_1 \cup E_2$, $|T| \leq 2r$.

For any $\gamma \in Z$, let p be a closest explaining codeword. The codeword $p - (c_0 + \gamma c_1)$ is supported on at most $3r \leq d - 1$ positions and is zero. Thus $\text{wt}(e_0 + \gamma e_1) \leq r$. Let T' be the support of the error pair. If $|T'| \geq r + 1$, every $\gamma \in Z$ makes at least $|T'| - r$ nonzero affine coordinate functions $e_{0,x} + \gamma e_{1,x}$ vanish. Each coordinate has at most one root, so $|Z|(|T'| - r) \leq |T'|$ and $|Z| \leq r + 1$.

It remains that $|T'| \leq r$. If γ is MCA-bad on a set S , $|S| \geq n - r$, its explaining codeword equals $c_0 + \gamma c_1$, since their difference would otherwise have weight at most $2r \leq d - 1$. The set S must meet T' , and at a point of $S \cap T'$ the nonzero pair satisfies $e_{0,x} + \gamma e_{1,x} = 0$. Such a point determines γ , giving at most $|T'| \leq r$ bad slopes. $\square$

Recall that for a nonempty challenge set $\Gamma \subseteq \mathbb{F}$, $B_{C,\Gamma}^{\text{MCA}}(a)$ denotes the same maximum with slopes restricted to Γ .

The following “tangent floor” is an elementary family of neighboring coordinate-span intersections. The name does not assert a hidden differentiability or multiplicity condition.

Proposition 4.2 (Universal tangent floor). *For every Reed–Solomon code, every $a \geq k + 1$, and every nonempty $\Gamma \subseteq \mathbb{F}$,*

$$(3.6) \quad B_{C,\Gamma}^{\text{MCA}}(a) \geq \min\{|\Gamma|, n - a + 1\}.$$

Proof. Put $j = n - a$, $t = \min\{|\Gamma|, j + 1\}$, and choose distinct $\gamma_1, \dots, \gamma_t \in \Gamma$ and independent parity columns $h_{x_1}, \dots, h_{x_t}$. Define $s_1 = \sum_i h_{x_i}$ and $s_0 = -\sum_i \gamma_i h_{x_i}$, and lift them to received words. For $E_i = \{x_\ell : \ell \neq i\}$, $s_0 + \gamma_i s_1 \in V_{E_i}$, whereas $s_1 \notin V_{E_i}$. Proposition 3.1 makes every γ_i bad on $D \setminus E_i$, which has size at least a . $\square$

Corollary 4.3 (Exact deep numerator). *If $C = \text{RS}_{\mathbb{F}}(D, k)$, $1 \leq k < n$, $k + 1 \leq a \leq n$, $\emptyset \neq \Gamma \subseteq \mathbb{F}$, and $3(n - a) \leq n - k$, then*

$$B_{C,\Gamma}^{\text{MCA}}(a) = \min\{|\Gamma|, n - a + 1\}.$$

Proof. The upper bound is [Theorem 4.1](#), intersected with Γ . The hypotheses imply $a \geq k + 1$, so [Proposition 4.2](#) gives equality. $\square$

Remark 4.4 (The zero-radius normalization). Under the formal affine sampler, [Corollary 4.3](#) at $a = n$ gives $B_C^{\text{MCA}}(n) = 1$, and hence $e_{\text{MCA}}(C, 0) = 1/|\mathbb{F}|$ for every proper Reed–Solomon code. This also follows directly: in $\mathbb{F}^D/C$, a noncontained affine line meets the zero coset in at most one parameter, and the bound is attained by a line through the zero coset. A value $2/|\mathbb{F}|$ belongs to a different or loose convention and must not be imported into the affine definition of [\[ABF26\]](#).

The next lemma isolates the deterministic mechanism behind both extensions of the deep range.

Lemma 4.5 (Large-overlap collapse). *Fix a received pair and an exact agreement $a = n - r \geq k + 1$. Choose one noncommon exact witness support S_z and explaining polynomial p_z for every slope z in its bad set Z . If two distinct chosen supports satisfy*

$$(MO1) \quad |S_z \cap S_w| \geq k + r,$$

then $|Z| \leq r + 1$.

Proof. Put

$$c_1 = \frac{p_z - p_w}{z - w}, \quad c_0 = p_z - zc_1.$$

On $S_z \cap S_w$, the two line equations imply that the received pair equals (c_0, c_1) . Let C_0 be the full common support of this codeword pair and write $c = |C_0| \geq k + r$. For every $u \in Z$, $|S_u \cap C_0| \geq a + c - n \geq k$, so the explaining polynomial p_u equals $c_0 + uc_1$.

Subtract this codeword pair from the received pair. At every coordinate outside C_0 at least one of the two residual values is nonzero, and a coordinate can cancel at at most one finite slope. A noncommon witness for u must contain at least $\max\{1, a - c\}$ such cancellations: the support-size condition gives $a - c$, and one is still necessary when $a \leq c$. Double counting therefore gives

$$(MO2) \quad |Z| \leq \left\lfloor \frac{n - c}{\max\{1, a - c\}} \right\rfloor \leq r + 1.$$

Indeed, for $c \geq a$ the numerator is at most r ; for $c = a - 1$ it is $r + 1$; and, with $x = a - c \geq 2$, the quotient is $(r + x)/x \leq r + 1$. $\square$

Theorem 4.6 (Quadratic mean-overlap theorem). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, put $R = n - k$, let $0 \leq r \leq R - 1$, and let $\emptyset \neq \Gamma \subseteq \mathbb{F}$. If*

$$(MO3) \quad (n - r)^2 \geq n(k + r), \quad \text{equivalently} \quad r^2 \geq n(3r - R),$$

then

$$(MO4) \quad B_{C, \Gamma}^{\text{MCA}}(n - r) = \min\{|\Gamma|, r + 1\}.$$

The same upper bound holds for every linear $[n, k]$ MDS code.

Proof. The universal tangent construction gives the lower bound. For the upper bound, fix a received pair and its challenge-restricted bad set Z . If $|Z| \leq r + 1$ there is nothing to prove. Otherwise choose $r + 2$ slopes and distinct exact witness supports $S_1, \dots, S_{r+2}$, each of size $a = n - r$. Suppose every pair had intersection at most $k + r - 1$, and put $d_x = \#\{i : x \in S_i\}$. Then

$$(MO5) \quad \sum_x \binom{d_x}{2} \leq \binom{r + 2}{2} (k + r - 1),$$

whereas Cauchy–Schwarz and $a^2/n \geq k + r$ give

$$(MO6) \quad \sum_x \binom{d_x}{2} \geq \frac{(r + 2)^2 a^2 / n - (r + 2)a}{2} \geq \frac{(r + 2)^2 (k + r) - (r + 2)a}{2}.$$

The last lower bound exceeds the upper bound in (MO5) by

$$\frac{r+2}{2}(k+3r-n+1).$$

When $3r > R$ this is positive. When $3r \leq R$, the exact deep theorem already gives the desired upper bound. Thus in the remaining case some pair satisfies (MO1), and Lemma 4.5 contradicts $|Z| \geq r+2$. The exact-support reduction and large-overlap lemma use only the MDS information-set property, proving the stated extension. $\square$

Corollary 4.7 (Exact quadratic staircase). *Put*

$$(MO7) \quad r_{\text{quad}}(n, k) = \left\lfloor \frac{3n - \sqrt{n(5n+4k)}}{2} \right\rfloor.$$

For every $0 \leq r \leq r_{\text{quad}}(n, k)$ and every nonempty challenge set Γ ,

$$B_{C, \Gamma}^{\text{MCA}}(n-r) = \min\{|\Gamma|, r+1\}.$$

If $k/n \rightarrow \rho$, then $r_{\text{quad}}(n, k)/n \rightarrow \alpha_\rho = (3 - \sqrt{5+4\rho})/2$.

Proof. The smaller root of $r^2 - 3nr + n(n-k) = 0$ is the real number inside the floor in (MO7). Apply Theorem 4.6; the limiting formula follows by division by n . $\square$

Proposition 4.8 (Limit of pairwise-overlap information). *Let $k/n \rightarrow \rho \in (0, 1)$, $r/n \rightarrow \alpha \in (0, 1)$, and suppose*

$$(MO8) \quad \alpha^2 < 3\alpha - (1 - \rho).$$

For all sufficiently large n , there are $r+2$ distinct r -subsets T_i such that, whenever $j = 3r - (n-k) \leq r$, $|T_i \cap T_\ell| \leq j-1$ for every $i \neq \ell$. If $j > r$, that intersection condition is automatic. Hence pairwise support overlap alone cannot extend the exact theorem past the quadratic boundary; further progress must use the syndrome, determinant, or fiber incidence. This is a limitation of the method, not a construction of an MCA-bad line.

Proof. When $j \leq r$, choose $r+2$ independent uniform r -subsets. The intersection of a fixed pair is hypergeometric with mean $r^2/n = (\alpha^2 + o(1))n$, whereas $j = (3\alpha - (1 - \rho) + o(1))n$. More explicitly,

$$\Pr(|T_1 \cap T_2| = s) = \frac{\binom{r}{s} \binom{n-r}{r-s}}{\binom{n}{r}}.$$

After writing $s = xn$, Stirling's formula gives the exponent

$$\alpha h(x/\alpha) + (1-\alpha)h((\alpha-x)/(1-\alpha)) - h(\alpha),$$

which is strictly concave and has its unique maximum zero at $x = \alpha^2$. The strict gap in (MO8), followed by summing at most $n+1$ terms, therefore gives probability $e^{-\Omega(n)}$ that one fixed intersection reaches j . A union bound over $O(n^2)$ pairs is smaller than one. A successful family is automatically distinct, because equal members have intersection $r \geq j$. When $j > r$, choose any $r+2$ distinct r -sets. $\square$

The following complementary theorem can be stronger when the overlap defect $j = 3r - R$ is small. Its only new input is a packing inequality; in particular it uses no smoothness or Fourier estimate.

Theorem 4.9 (Exact overlap-packing criterion). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, put $R = n - k$, and fix an integer $0 \leq r \leq R - 1$. Set $a = n - r$ and $j = 3r - R$. If either*

$$(OP0) \quad j \leq 0,$$

or

$$(OP) \quad 1 \leq j \leq r, \quad \binom{n}{j} \leq (r+1) \binom{r}{j},$$

then

$$(OP') \quad B_C^{\text{MCA}}(n - r) = r + 1.$$

The same upper bound holds for every linear $[n, k]$ MDS code; the matching lower bound holds whenever the field contains at least $r + 1$ distinct slope parameters.

Proof. Condition (OP0) is [Corollary 4.3](#), so assume (OP). Fix an arbitrary received pair (f, g) , let Z be its bad-slope set, and use [Lemma 2.10](#) to choose, for every $z \in Z$, an exact witness support S_z , $|S_z| = a$, and an explaining polynomial $p_z \in \mathbb{F}[X]_{<k}$. A single noncommon support cannot carry two distinct slopes, so the supports chosen for distinct elements of Z are distinct.

For the stated MDS extension, the same exact-support reduction is available: interpolate the unexplained component on any k -subset of the original witness support; because every k -subset is an information set, nonexplanation is detected at one further coordinate, and that $(k + 1)$ -set can be enlarged to size a .

First suppose that two of them have a large intersection:

$$(OP1) \quad |S_z \cap S_w| \geq n + k - a = k + r \quad (z \neq w).$$

Then [Lemma 4.5](#) gives $|Z| \leq r + 1$.

It remains to consider the complementary case, in which every pair of chosen supports satisfies $|S_z \cap S_w| \leq n + k - a - 1$. Put $T_z = D \setminus S_z$, so $|T_z| = r$. For distinct slopes,

$$(OP3) \quad |T_z \cap T_w| = n - 2a + |S_z \cap S_w| \leq 3r - R - 1 = j - 1.$$

No j -subset of D can therefore lie in two different T_z 's. Counting pairs (z, J) with $J \in \binom{T_z}{j}$ yields

$$|Z| \binom{r}{j} \leq \binom{n}{j} \leq (r + 1) \binom{r}{j},$$

and hence $|Z| \leq r + 1$. The universal tangent construction gives the reverse inequality. The proof used only linearity, the fact that agreement of two codewords on k coordinates forces equality, and the exact-support reduction; these are MDS properties. The coordinate tangent construction gives the stated MDS lower bound. $\square$

Proposition 4.10 (General overlap–packing upper bound). *For $1 \leq j = 3r - (n - k) \leq r$, let $A(n, r, j)$ be the largest size of a family of r -subsets of an n -set whose distinct members intersect in at most $j - 1$ points. Then*

$$(OP5) \quad B_{C,\Gamma}^{\text{MCA}}(n - r) \leq \min \{|\Gamma|, \max(r + 1, A(n, r, j))\}$$

and

$$(OP6) \quad A(n, r, j) \leq \left\lfloor \frac{\binom{n}{j}}{\binom{r}{j}} \right\rfloor.$$

Proof. For a fixed received pair, either two chosen exact witness supports have intersection at least $k + r$, in which case [Lemma 4.5](#) gives at most $r + 1$ slopes, or every pair of complementary r -sets intersects in at most $j - 1$ points. The latter family has size at most $A(n, r, j)$. Finally, no j -subset can lie in two members of such a packing, so counting contained j -subsets proves [\(OP6\)](#). $\square$

Corollary 4.11 (Quadratic staircase at the official rates). *Let $n = 2^e$, let $\rho = k/n \in \{1/2, 1/4, 1/8, 1/16\}$, and put*

$$r_\rho(n) = \left\lfloor n \frac{3 - \sqrt{5 + 4\rho}}{2} \right\rfloor.$$

For every n -point Reed–Solomon domain over every field,

$$(OP4) \quad B_C^{\text{MCA}}(n - r) = r + 1 \quad (0 \leq r \leq r_\rho(n)).$$

Proof. This is [Corollary 4.7](#), since the displayed integer is $r_{\text{quad}}(n, k)$. $\square$

Corollary 4.12 (Self-contained quadratic-range target compiler). *In the setting of Corollary 4.11, put $q = |\mathbb{F}|$ and $B = \lfloor q/2^{128} \rfloor$. If*

$$1 \leq B \leq M_\rho(n) := \min\{r_\rho(n) + 1, n - k - 1\},$$

then

$$(PW) \quad e_{MCA}(C, \delta) \leq 2^{-128} \iff \lfloor \delta n \rfloor \leq B - 1.$$

Thus every such row is solved throughout the field window $2^{128} \leq q < 2^{128}(M_\rho(n) + 1)$: its largest safe grid radius is $(B - 1)/n$, while its real transition supremum is B/n and that endpoint is unsafe. If $q < 2^{128}$, even radius zero is unsafe.

Proof. For $r \leq B - 1$, Corollary 4.11 gives the exact numerator $r + 1 \leq B \leq q/2^{128}$. At $r = B$, the tangent floor applies because $B \leq n - k - 1$, and gives at least $B + 1 > q/2^{128}$ slopes. Monotonicity proves the equivalence and the endpoint statements. If $B = 0$, the tangent floor at $r = 0$ gives the exact numerator one. $\square$

5. LITERATURE COMPLETION TO POSITIVE RELATIVE RADIUS

The quadratic theorem is self-contained and useful at finite parameters, but it is not the largest known RS range at constant rate. We now make the relevant implication of [BCHKS25] explicit at the integer numerator level. This section is deliberately separated from Section 4: the input below belongs to the literature, whereas Theorem 4.6 is the finite MDS argument of this paper.

Put $q = |\mathbb{F}|$. For a word u and a received pair (u_0, u_1) , write $d(u, C)$ and $d_2((u_0, u_1), C^2)$ for their relative Hamming and relative column distances from C and C^2 , respectively. For $0 \leq \alpha < \beta < 1$, define the two-radius correlated-agreement error by

$$(LC0) \quad e_{CA}(C, \alpha, \beta) = \frac{1}{q} \max_{\substack{u_0, u_1 \in \mathbb{F}^D \\ d_2((u_0, u_1), C^2) > \beta}} \#\{\gamma \in \mathbb{F} : d(u_0 + \gamma u_1, C) \leq \alpha\}.$$

The one-radius notation $e_{CA}(C, \alpha)$ means that the far-pair radius is also α .

Theorem 5.1 (BCHKS two-radius bound). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, put $q = |\mathbb{F}|$, $\rho = k/n$, and $\Delta = (n - k + 1)/n$. If $\Delta/3 \leq \alpha < (1 - \rho)/2$ and $\alpha < \beta < 1$, then*

$$(BCHKS) \quad e_{CA}(C, \alpha, \beta) \leq \frac{1}{q} \max \left\{ \frac{1 - \rho - \alpha}{\alpha(1 - \rho - 2\alpha)}, \frac{\beta}{\beta - \alpha} \right\}.$$

This is [BCHKS25, Thm. 1.3], in the two-radius normalization restated as [ABF26, Thm. 4.9(2)]. We have written only the strict subrange needed below, so all displayed denominators are positive.

Theorem 5.2 (Integral BCHKS completion). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, let $q = |\mathbb{F}|$, put $R = n - k$, and let $r \in \mathbb{Z}$ satisfy*

$$(LC1) \quad 3r \geq R + 1, \quad 2r < R.$$

Then

$$(LC2) \quad B_C^{\text{MCA}}(n - r) \leq \left\lfloor \max \left\{ \frac{n(R - r)}{r(R - 2r)}, r + 1 \right\} \right\rfloor.$$

Consequently, if

$$(LC3) \quad n(R - r) < r(r + 2)(R - 2r),$$

then, for every nonempty $\Gamma \subseteq \mathbb{F}$,

$$(LC4) \quad B_{C, \Gamma}^{\text{MCA}}(n - r) = \min\{|\Gamma|, r + 1\}.$$

Proof. Put $\delta_{\text{fld}} = r/n$ and, for $0 < \theta < 1$, put $\delta_{\text{int}} = (r + \theta)/n$. Hamming distances belong to $\{0, 1/n, \dots, 1\}$, so replacing the far-pair radius δ_{fld} by δ_{int} does not change the far-pair condition; this is the grid observation of [ABF26, Rem. 4.10]. The hypotheses in (LC1) give $\Delta/3 \leq \delta_{\text{fld}} < (1 - k/n)/2$, and $\delta_{\text{fld}} < \delta_{\text{int}} < 1$. Hence Theorem 5.1 gives

$$(LC5) \quad qe_{\text{CA}}(C, r/n) \leq \max \left\{ \frac{n(R - r)}{r(R - 2r)}, \frac{r + \theta}{\theta} \right\}.$$

On the integer grid, $qe_{\text{CA}}(C, r/n) = B_C^{\text{CA}}(n - r)$. The second inequality in (LC1) also places the radius below $d_{\text{min}}/(2n)$, because $d_{\text{min}} = R + 1$. By Theorem 2.5,

$$B_C^{\text{MCA}}(n - r) = \max\{B_C^{\text{CA}}(n - r), \min\{q, r\}\}.$$

The right side of (LC5) is strictly larger than r , because $(r + \theta)/\theta > r$. It therefore bounds both branches of this maximum and hence bounds the integer $B_C^{\text{MCA}}(n - r)$. For the full-field maximum one may alternatively invoke the known below-half equality [ABF26, Lem. 4.6]; the sparse decomposition is kept here because it is exact before maximizing over a restricted challenge set.

Letting $\theta \rightarrow 1^-$ proves that this integer is at most the real maximum in (LC2), and taking the floor proves (LC2). Under (LC3) the first entry of that maximum is strictly less than $r + 2$. Since $r + 1$ is integral, the floor in (LC2) is at most $r + 1$, so the full-field numerator is at most $r + 1$. Restriction to Γ gives the upper bound $\min\{|\Gamma|, r + 1\}$, and Proposition 4.2 gives the reverse bound. $\square$

Remark 5.3 (Priority). The substance of the upper bound in Theorem 5.2 is prior work. In particular, [BCHKS25, Cor. 1.4] states a sharp lossless conclusion under broad finite hypotheses and [BCHKS25, Rem. 2.5] discusses the matching perturbation construction. Formula (LC5) is included to expose the integer limit and to make the challenge-subset and target compilers below completely transparent.

Corollary 5.4 (Positive density of exact radii). *Let $k_n/n \rightarrow \rho \in (0, 1)$, let $r_n \in \mathbb{Z}$ satisfy $r_n/n \rightarrow \tau$, and suppose $0 < \tau < (1 - \rho)/2$. For every RS row sequence and every sequence of nonempty challenge sets,*

$$(LC6) \quad B_{C_n, \Gamma_n}^{\text{MCA}}(n - r_n) = \min\{|\Gamma_n|, r_n + 1\}$$

for all sufficiently large n . More uniformly, for every $\eta > 0$, equality holds throughout

$$0 \leq r \leq \left(\frac{1 - \rho}{2} - \eta \right) n$$

for every sufficiently large row whose rate tends to ρ . Consequently the exact staircase contains an initial interval of normalized length $(1 - \rho)/2 - o(1)$, has lower density at least $(1 - \rho)/2$ in the full Hamming-radius interval, and has asymptotic relative density one in the unique-decoding interval.

Proof. The quadratic theorem covers every sequence whose limiting radius lies below $\alpha_\rho = (3 - \sqrt{5 + 4\rho})/2$. In the remaining compact subinterval of $(0, (1 - \rho)/2)$, the hypotheses (LC1) hold for large n , while

$$\frac{n(R - r)}{r(R - 2r)} = O_{\rho, \eta}(1) \quad \text{and} \quad r + 1 = \Theta_{\rho, \eta}(n).$$

Thus (LC3) holds and Theorem 5.2 applies. The uniform statement follows by the same compactness argument, using the deep theorem near radius zero. Counting integer radii gives the density assertions. $\square$

Corollary 5.5 (Official-rate staircase). *Let $n = 2^e \geq 128$, $k = \rho n$, and $\rho \in \{1/2, 1/4, 1/8, 1/16\}$. Define*

$$(LC7) \quad r_\rho^\# = \begin{cases} n/4 - 3, & \rho = 1/2, \\ 3n/8 - 2, & \rho = 1/4, \\ 7n/16 - 2, & \rho = 1/8, \\ 15n/32 - 2, & \rho = 1/16. \end{cases}$$

Then for every RS domain and every nonempty $\Gamma \subseteq \mathbb{F}$,

$$(LC8) \quad B_{C,\Gamma}^{\text{MCA}}(n-r) = \min\{|\Gamma|, r+1\} \quad (0 \leq r \leq r_\rho^\#).$$

Proof. Put $F(r) = r(r+1)(R-2r) - n(R-r)$, where $R = n-k$, and $L = \lceil (R+1)/3 \rceil$. The deep theorem covers $r < L$. On the positive real axis, $F'(r) = -6r^2 + 2(R-2)r + (R+n)$ has exactly one positive zero, so the minimum of F on $[L, r_\rho^\#]$ occurs at an endpoint. Since $R \geq n/2 \geq 64$,

$$F(L) \geq \frac{(R+1)(R+4)(R-8) - 9n(2R-1)}{27} > 0;$$

indeed the numerator is at least $R\{R(R-4) - 18n\} > 0$. Direct substitution at the other endpoint gives, in the order of the four rates,

$$\frac{n^2 - 84n + 288}{8}, \quad \frac{3n^2 - 104n + 128}{16}, \quad \frac{21n^2 - 464n + 512}{64}, \quad \frac{105n^2 - 1952n + 2048}{256},$$

all positive for $n \geq 128$. Therefore $F(r) \geq 0$ throughout the remaining integer interval. Since $r > 0$ and $R - 2r > 0$ there,

$$r(r+2)(R-2r) - n(R-r) = F(r) + r(R-2r) > 0,$$

which is (LC3). Apply Theorem 5.2. $\square$

Corollary 5.6 (Almost-half-distance target compiler). *In the setting of Corollary 5.5, put $B = \lfloor \varepsilon |\Gamma| \rfloor$. If*

$$1 \leq B < |\Gamma|, \quad B \leq r_\rho^\# + 1,$$

then the complete safe set is

$$(LC9) \quad \{\delta : e_{\text{MCA},\Gamma}(C, \delta) \leq \varepsilon\} = [0, B/n],$$

and the endpoint is unsafe. For the full-field target 2^{-128} , this applies throughout $2^{128} \leq q < 2^{128}(r_\rho^\# + 2)$.

Proof. At every radius $r \leq B-1$, Corollary 5.5 gives numerator $r+1 \leq B$. At radius B , the tangent floor gives at least $B+1$ bad slopes. Monotonicity and $r(\delta) = \lfloor \delta n \rfloor$ prove the assertion. $\square$

6. EXACT THRESHOLDS FOR EXPLICIT PRIZE ROWS

TABLE 1. Explicit maximal-dimension prize-admissible rows. In each row $k = 2^{40}$, $B = \lfloor p/2^{128} \rfloor = r_\rho(n) + 1$, and (s, a_0) is part of the Proth certificate in Proposition 6.1.

rate	length	B	bits of p	(s, a_0)
1/2	2^{41}	389500552609	167	(92, 3)
1/4	2^{42}	1210584858040	169	(93, 13)
1/8	2^{43}	2879806199253	170	(95, 5)
1/16	2^{44}	6233898019554	171	(97, 5)

For completeness, the four field orders are

$$\begin{aligned} p_{1/2} &= 132540169958804033333249306710494641010898987122689, \\ p_{1/4} &= 411940680852499481698306614369841346700408394874881, \\ p_{1/8} &= 979947269755402568812854322316630667196565607677953, \\ p_{1/16} &= 2121285573237585848299875619011192262679065433997313. \end{aligned}$$

Their odd Proth coefficients u , in the same order, are

$$\begin{aligned} u_{1/2} &= 26766274163673319604503, & u_{1/4} &= 41595378994516821279015, \\ u_{1/8} &= 24737346889219389259839, & u_{1/16} &= 13387194060291799253121. \end{aligned}$$

For $F_{n,k}(r) = r^2 - n(3r - (n - k))$, the exact boundary checks are

	ρ	$F_{n,k}(B - 1)$	$F_{n,k}(B)$
(PC0)	1/2	5154112775168	-663955886271
	1/4	7590647904465	-3182321912768
	1/8	13908181940112	-6720484728007
	1/16	19335616403905	-20973145690236.

Proposition 6.1 (Exact arithmetic and primality certificates). *For the four rows in Table 1, the displayed integer p satisfies*

$$(PC1) \quad p = u2^s + 1, \quad u \text{ odd}, \quad u < 2^s, \quad a_0^{(p-1)/2} \equiv -1 \pmod{p}.$$

Consequently every p is prime. Moreover

$$(PC2) \quad n \mid p - 1, \quad p < 2^{256}, \quad B2^{128} \leq p < (B + 1)2^{128},$$

where n and B are the corresponding entries of the table. They also satisfy $F_{n,k}(B - 1) \geq 0 > F_{n,k}(B)$. Since $F'_{n,k}(r) = 2r - 3n < 0$ on $[0, n]$, these two signs locate its smaller root and give $B - 1 = r_\rho(n) = r_{\text{quad}}(n, k)$. Thus $\mathbb{F}_p^*$ contains a subgroup of order n , and each code in Theorem 1.4 exists with the asserted slope budget.

Proof. All identities and inequalities in (PC0)–(PC2) are direct integer calculations; the values of (s, a_0) , u , p , n , and B have all been printed above so that the calculation is independently reproducible. We recall why the certificate proves primality. If a prime ℓ divides p , then $a_0^{u2^{s-1}} \equiv -1 \pmod{\ell}$, whereas $a_0^{u2^s} \equiv 1 \pmod{\ell}$. The multiplicative order of a_0 modulo ℓ therefore has exact 2-part 2^s , so $2^s \mid \ell - 1$. Since $u < 2^s$, one has $2^s > \sqrt{p}$; hence a composite p would have a prime divisor both at most $\sqrt{p}$ and at least $2^s + 1$, a contradiction. This is Proth's theorem in the special form used here. Finally $s \geq \log_2 n$ gives $n \mid p - 1$, and cyclicity of $\mathbb{F}_p^*$ gives the required subgroup. $\square$

Theorem 6.2 (Exact first adjacent row over a separating field). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, put $R = n - k$ and $M = \binom{n}{R-1} = \binom{n}{k+1}$, and define*

$$Q_{\text{sep}}(M) = \max \left\{ M, \binom{M}{2} \right\}.$$

If $|\mathbb{F}| > Q_{\text{sep}}(M)$, then for the full-field challenge

$$(AD1) \quad B_C^{\text{MCA}}(k + 1) = M.$$

If $R \geq 2$ and $b = \lfloor \varepsilon |\mathbb{F}| \rfloor$, the exact target threshold therefore satisfies

$$\begin{aligned} (AD2) \quad b \geq M &\implies a^* = k + 1, \\ \binom{n}{R-2} \leq b < M &\implies a^* = k + 2. \end{aligned}$$

Thus the first finite adjacent comparison has literal binomial constants; no $e^{o(n)}$ or unspecified polynomial factor occurs.

Proof. The upper bound in (AD1) is [Proposition 2.11](#). For the reverse inequality use the Reed–Solomon parity check of redundancy R . For every $(R - 1)$ -set $E \subseteq D$, its column span V_E is a hyperplane in $\mathbb{F}^R$, and distinct sets give distinct hyperplanes: otherwise the union of two distinct $(R - 1)$ -sets would contain at least R columns in one $(R - 1)$ -space, contradicting the MDS column property.

Every proper linear hyperplane in $\mathbb{F}^R$ has $|\mathbb{F}|^{R-1}$ points. Since $M < |\mathbb{F}|$, the union of the M spaces V_E has fewer than $|\mathbb{F}|^R$ points; choose y_1 outside it. For each E , choose a nonzero linear functional ℓ_E with kernel V_E . The line $y_0 + \gamma y_1$ meets V_E at

$$\gamma_E = -\ell_E(y_0)/\ell_E(y_1).$$

For $E \neq E'$, equality $\gamma_E = \gamma_{E'}$ cuts out a proper hyperplane in the variable y_0 , because the normalized functionals $\ell_E/\ell_E(y_1)$ and $\ell_{E'}/\ell_{E'}(y_1)$ are distinct. There are at most $\binom{M}{2} < |\mathbb{F}|$ collision hyperplanes; their union also has fewer than $|\mathbb{F}|^R$ points, so choose y_0 outside it. Since $y_1 \notin V_E$, the line is not contained in V_E , and every intersection is transverse. The M parameters γ_E are then finite and pairwise distinct. Syndrome surjectivity and [Theorem 3.2](#) realize them as M bad slopes at agreement $k + 1$, proving (AD1).

If $b \geq M$, the support upper bound makes the first allowed agreement safe. If $\binom{n}{R-2} \leq b < M$, equality (AD1) makes $k + 1$ unsafe, whereas [Proposition 2.11](#) makes $k + 2$ safe. Monotonicity proves (AD2). $\square$

Theorem 6.3 (Subfield confinement). *Let $\mathbb{B} \subseteq \mathbb{F}$, $D \subseteq \mathbb{B}$, and let $u_0, u_1 \in \mathbb{B}^D$. If $\gamma \in \mathbb{F} \setminus \mathbb{B}$ and $u_0 + \gamma u_1$ is explained on S , then the pair is explained on the same S . Hence every MCA-bad slope of a $\mathbb{B}$ -valued received line lies in $\mathbb{B}$.*

Proof. Choose a $\mathbb{B}$ -basis $1, \beta_1, \dots$ of $\mathbb{F}$. Decompose the coefficients of the explaining polynomial and of γ in this basis. Because $D \subseteq \mathbb{B}$, comparison of basis components on S gives a codeword explaining u_1 from any nonzero nonbase component of γ , and then a codeword explaining u_0 from the base component. $\square$

7. EXACT LOCATOR PREFIXES AND COLLISION-AWARE POLE LINES

Assume $D \subseteq \mathbb{B} \subseteq \mathbb{F}$. For an m -set $S \subseteq D$, write

$$Q_S(X) = \prod_{x \in S} (X - x) = X^m + c_1(S)X^{m-1} + \dots + c_m(S), \quad \Phi_w(S) = (c_1(S), \dots, c_w(S)).$$

The monic polynomial Q_S , whose simple roots are precisely the points of S , is the *support locator*. Its *depth- w locator prefix* is the vector of the first w coefficients below the leading coefficient. For $z \in \mathbb{B}^w$, the inverse image $\Phi_w^{-1}(z)$ is the *prefix fiber over z* : the family of all m -element supports having that same prefix. Thus “fiber” in this paper always refers to the inverse image of a displayed map.

Proposition 7.1 (Exact prefix-list bijection). *Let $1 \leq K \leq m \leq n$, put $w = m - K$, and set $U_z(X) = X^m + \sum_{i=1}^w z_i X^{m-i}$ for $z \in \mathbb{B}^w$. The codewords of $\text{RS}_{\mathbb{F}}(D, K)$ agreeing with $U_z|_D$ on at least m points are exactly*

$$(4.1) \quad (U_z - Q_S)|_D, \quad S \in \Phi_w^{-1}(z).$$

No such codeword agrees on more than m points. In particular, some received word has list size at least $\lceil \binom{n}{m} |\mathbb{B}|^{-w} \rceil$.

Proof. For $S \in \Phi_w^{-1}(z)$, the leading term and next w coefficients cancel, so $\deg(U_z - Q_S) < m - w = K$. Conversely, if P , $\deg P < K$, agrees with U_z at m points, then $U_z - P$ is monic of degree m with those m roots and equals their locator. Coefficient comparison gives the prefix. A nonzero degree- m polynomial has no additional root. Pigeonhole the $\binom{n}{m}$ supports over $|\mathbb{B}|^w$ prefixes. $\square$

The *simple-pole conversion* chooses an auxiliary point $\alpha \notin D$ and divides coordinatewise by $X - \alpha$. This turns one list of nearby polynomials into points on a received affine line for the

dimension- k code. A *collision* is an equality $P_i(\alpha) = P_j(\alpha)$ for two distinct listed polynomials; the next formula pays such equalities rather than assuming all slopes are distinct.

Theorem 7.2 (Collision-aware simple-pole conversion). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, $C^+ = \text{RS}_{\mathbb{F}}(D, k + 1)$, $1 \leq k < n$, and $q = |\mathbb{F}| > n$. If a word has L distinct codewords of C^+ agreeing on at least $m \geq k + 1$ points, then one received line for C has at least*

$$(4.2) \quad M(L) = \left\lceil \frac{L(q - n)}{q - n + k(L - 1)} \right\rceil$$

distinct MCA-bad slopes at agreement m .

Proof. Let the received word be $U \in \mathbb{F}^D$, and let the listed polynomials be $P_1, \dots, P_L$, each of degree at most k . For $\alpha \in \mathbb{F} \setminus D$, define the word coordinatewise by $f_\alpha(x) = U(x)/(x - \alpha)$ and set $g_\alpha(x) = -1/(x - \alpha)$. On an agreement set for P_i , the line point of slope $P_i(\alpha)$ equals $(P_i(x) - P_i(\alpha))/(x - \alpha)$, a degree-less-than- k evaluation. The pair is not simultaneously explained on $k + 1$ points: an explanation of g_α by G , $\deg G < k$, would make $(X - \alpha)G(X) + 1$ vanish at $k + 1$ points although its degree is at most k and its value at α is 1.

For $i \neq j$, the equality $P_i(\alpha) = P_j(\alpha)$ holds at at most k poles. Some pole has at most $k \binom{L}{2} / (q - n)$ colliding pairs. If its value multiplicities are $m_1, \dots, m_M$, then $\sum m_i^2 \leq L + kL(L - 1)/(q - n)$. Cauchy–Schwarz gives $L^2 \leq M \sum m_i^2$, which yields (4.2). $\square$

Corollary 7.3 (Exact good-pole reservoir). *In Theorem 7.2, let $Q = q - n$, and for $\alpha \in \mathbb{F} \setminus D$ let C_α be the number of unordered pairs $\{i, j\}$ with $P_i(\alpha) = P_j(\alpha)$. For every integer $T \geq 0$, at least*

$$(PR1) \quad \max \left\{ 0, Q - \left\lfloor \frac{k \binom{L}{2}}{T + 1} \right\rfloor \right\}$$

poles satisfy $C_\alpha \leq T$, and every such pole produces at least

$$(PR2) \quad \left\lfloor \frac{L^2}{L + 2T} \right\rfloor$$

distinct MCA-bad slopes. In particular at least $\max\{0, Q - k \binom{L}{2}\}$ poles are fully separating.

Proof. Every nonzero difference $P_i - P_j$ has at most k roots, so

$$\sum_{\alpha \in \mathbb{F} \setminus D} C_\alpha \leq k \binom{L}{2}.$$

Every pole with $C_\alpha > T$ contributes at least $T + 1$ to this sum, which gives (PR1). At a good pole let $m_1, \dots, m_M$ be the multiplicities of the distinct list values. Then $\sum_i m_i = L$ and $\sum_i m_i^2 = L + 2C_\alpha \leq L + 2T$. Cauchy–Schwarz yields $M \geq L^2 / (L + 2T)$, proving (PR2); $T = 0$ gives the last assertion. $\square$

Remark 7.4 (One pole, one received line). The reservoir in Corollary 7.3 fixes the same pole when counting collisions, because different poles define different received lines. A cross-pole equation $P_i(\alpha) = P_j(\alpha')$ is one equation in two variables and generically defines a curve; a Bezout degree bound does not turn it into a finite collision set. Such cross-line collisions are neither controlled by the simple-pole argument nor needed by it.

Corollary 7.5 (Identity-prefix MCA floor). *For $m \geq k + 1$, put $w = m - k - 1$ and $L_m = \lceil \binom{n}{m} |\mathbb{B}|^{-w} \rceil$. If $q > n$, then*

$$(4.3) \quad B_C^{\text{MCA}}(m) \geq M(L_m).$$

For equal-size sets, the *Johnson distance* is $d_J(S, T) = |S \setminus T| = |T \setminus S|$. It counts the number of point replacements required to turn one support into the other.

Proposition 7.6 (Prefix rigidity and its packing limit). *Two distinct m -sets in one depth- w prefix fiber have Johnson distance at least $w + 1$. Consequently, with $t = \lfloor w/2 \rfloor$, every fiber has size at most*

$$(4.4) \quad \frac{\binom{n}{m}}{\sum_{i=0}^t \binom{m}{i} \binom{n-m}{i}}.$$

At $w \asymp n/\log|\mathbb{B}|$, this packing loss is only $e^{o(n)}$ and does not supply the random factor $|\mathbb{B}|^{-w}$.

Prefix rigidity says that sharing many leading locator coefficients forces distinct supports to be far apart in this metric. Here the *polynomial-field regime* means $|\mathbb{B}| = n^{O(1)}$, equivalently $\log|\mathbb{B}| = O(\log n)$.

Proof. If S, T have Johnson distance $j \leq w$, factor the locator of their intersection, of degree $m - j$, from $Q_S - Q_T$. The residual factor is nonzero, so the difference has degree at least $m - j \geq m - w$, whereas the common prefix makes its degree at most $m - w - 1$, a contradiction. Radius t Johnson balls about the fiber members are disjoint and each has the denominator in (4.4). For $w = n/\log|\mathbb{B}|$, the logarithm of that ball volume is $O(w \log(n/w)) = o(n)$ in the polynomial-field regime. $\square$

Theorem 7.7 (Prefix fibers embed in received lines). *Let $\mathcal{G} \subseteq \Phi_w^{-1}(z)$ be a fiber of N many m -sets, with $0 \leq w \leq m - 2$, and put $k = m - w - 1$. Every finite extension $\mathbb{F}/\mathbb{B}$ satisfying*

$$(4.5) \quad |\mathbb{F}| - n > k \binom{N}{2}$$

admits one received line for $\text{RS}_{\mathbb{F}}(D, k)$ with at least N distinct MCA-bad slopes furnished by $\mathcal{G}$.

Proof. For distinct $S, T \in \mathcal{G}$, the polynomial $Q_S - Q_T$ is nonzero of degree at most k . Condition (4.5) leaves a pole $\alpha \in \mathbb{F} \setminus D$ avoiding the roots of all these differences. Thus the values $Q_S(\alpha)$ are pairwise distinct. By Proposition 7.1, the polynomials $U_z - Q_S$ form one list for the dimension- $k + 1$ code; the pole line makes their values, and hence the resulting MCA slopes, pairwise distinct. $\square$

Theorem 7.8 (Exact list–line bijection at a separating pole). *Let $D \subseteq \mathbb{B} \subseteq \mathbb{F}$, let $1 \leq k < n$, let $m \geq k + 1$, and let $U \in \mathbb{B}^D$. Write*

$$\mathcal{L}_m(U) = \{P \in \mathbb{B}[X] : \deg P \leq k, |\{x \in D : U(x) = P(x)\}| \geq m\}.$$

Suppose $\alpha \in \mathbb{F} \setminus D$ and the evaluation map $P \mapsto P(\alpha)$ is injective on $\mathcal{L}_m(U)$. Define

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \quad g_\alpha(x) = -\frac{1}{x - \alpha} \quad (x \in D).$$

Then $P \mapsto P(\alpha)$ is a bijection from $\mathcal{L}_m(U)$ onto the set of finite MCA-bad slopes of the received line (f_α, g_α) for $\text{RS}_{\mathbb{F}}(D, k)$ at agreement m . The corresponding explaining polynomial is

$$h_P(X) = \frac{P(X) - P(\alpha)}{X - \alpha},$$

and its agreement set with $f_\alpha + P(\alpha)g_\alpha$ is exactly the agreement set of P with U .

If $L = |\mathcal{L}_m(U)|$, such a pole exists in every finite extension satisfying

$$(4.6) \quad |\mathbb{F}| > n + k \binom{L}{2}.$$

Proof. For $P \in \mathcal{L}_m(U)$, the polynomial h_P has degree less than k , and on every point where $U = P$ one has

$$f_\alpha + P(\alpha)g_\alpha = \frac{P - P(\alpha)}{X - \alpha} = h_P.$$

The word g_α cannot be explained by a polynomial G of degree less than k on $k + 1$ points, since then $(X - \alpha)G + 1$ would have degree at most k , at least $k + 1$ roots, and value one at α . Thus every displayed slope is MCA-bad.

Conversely, suppose $f_\alpha + \gamma g_\alpha$ is explained by $h \in \mathbb{F}[X]_{<k}$ on at least m points. Then $P(X) = (X - \alpha)h(X) + \gamma$ has degree at most k and agrees with U there. Interpolation on any $k+1$ of those points uses a Vandermonde system over $\mathbb{B}$ with right-hand side in $\mathbb{B}$, so $P \in \mathbb{B}[X]$. Hence $P \in \mathcal{L}_m(U)$ and $\gamma = P(\alpha)$. Injectivity proves the bijection, and the same calculation proves equality of the agreement sets.

For distinct $P, Q \in \mathcal{L}_m(U)$, the nonzero polynomial $P - Q$ has degree at most k , and therefore at most k roots. The union of D and the root sets of all $\binom{L}{2}$ differences has size at most $n + k\binom{L}{2}$; a larger extension contains an admissible pole. $\square$

Corollary 7.9 (Exact prefix-fiber ray realization). *Let $0 \leq w \leq m - 2$, put $k = m - w - 1$, and let $\mathcal{F}_z = \Phi_w^{-1}(z)$ be a complete depth- w fiber of m -subsets of D . If α separates the values $Q_S(\alpha)$, $S \in \mathcal{F}_z$, then the pole line associated with $U_z = X^m + \sum_{i=1}^w z_i X^{m-i}$ has exactly $|\mathcal{F}_z|$ MCA-bad slopes at agreement m . The maps*

$$S \mapsto U_z(\alpha) - Q_S(\alpha) \mapsto \left(U_z(\alpha) - Q_S(\alpha), S, \frac{U_z - Q_S - (U_z(\alpha) - Q_S(\alpha))}{X - \alpha} \right)$$

are bijections between the prefix fiber, the bad-slope set, and the exact- m witness incidence. In particular, the complete prefix fiber, its bad-slope set, and its exact-agreement witness incidence have the same cardinality.

Proof. The exact prefix-list bijection identifies $\mathcal{F}_z$ with the complete dimension- $(k+1)$ list for U_z . Apply Theorem 7.8. Since $U_z - (U_z - Q_S) = Q_S$ has precisely the roots in S , every agreement is exactly m and supplies exactly that support. $\square$

8. FURTHER EXACT CONSTRUCTIONS

The main threshold theorem needs only the syndrome correspondence and the mean-overlap argument. This section records complementary exact constructions that clarify the support-to-slope projection and the structured domains used in the examples. Every statement is a finite identity, inequality, or construction; no equidistribution or inverse theorem is assumed.

8.1. Determinantal ray normalization. The final projection from supports to slopes can be made explicit without postulating a support-pair multiplicity. The construction below applies to every Reed–Solomon row, in every characteristic. It also identifies the exact additional boundary maps that would govern extensions beyond the proved overlap range. No estimate for those maps is used in the main threshold theorem.

The folding atlas is stated for agreement supports $S \in \binom{D}{a}$, whereas this subsection uses their complementary error supports $E = D \setminus S \in \binom{D}{n-a}$. Complementation is a bijection. On a degree- c complete fold it sends a fiber occupied in j agreement points to one occupied in $c - j$ error points, so $\nu_{\pi,j}^E = \nu_{\pi,c-j}^S$; it also complements the recorded subset of the bounded exceptional set. Hence the joint-signature census and its exact support add-back transfer to the error-support slices without loss.

Let $C = \text{RS}_{\mathbb{F}}(D, k)$, put $R = n - k$, and use the weighted Vandermonde parity-check columns

$$h_x = \lambda_x(1, x, \dots, x^{R-1})^\top, \quad \lambda_x \neq 0,$$

as in the syndrome–secant normal form. For $E \subseteq D$, write $V_E = \text{Span}\{h_x : x \in E\}$. Fix an exact agreement $a \geq k + 1$, and set

$$(DR0) \quad t = n - a, \quad d = R - t = a - k.$$

In particular, $d - 1 = a - k - 1$ is precisely the identity locator-prefix depth. For $E \in \binom{D}{t}$, write its error locator in ascending order as

$$Q_E(X) = \prod_{x \in E} (X - x) = \sum_{\ell=0}^t q_\ell(E) X^\ell, \quad q_t(E) = 1.$$

For $y = (\eta_0, \dots, \eta_{R-1}) \in \mathbb{F}^R$, define the *locator-recurrence boundary*

$$(DR1) \quad \mathcal{A}_E(y) = \left(\sum_{\ell=0}^t q_\ell(E) \eta_{j+\ell} \right)_{0 \leq j < d} \in \mathbb{F}^d.$$

If $f(X) = \sum_{i=0}^{R-1} f_i X^i$, we use the coefficient pairing $\langle y, f \rangle = \sum_{i=0}^{R-1} y_i f_i$.

Lemma 8.1 (Locator recurrence and a secant span). *For every $E \in \binom{D}{t}$ and $y \in \mathbb{F}^R$,*

$$(DR2) \quad y \in V_E \iff \mathcal{A}_E(y) = 0.$$

Proof. Identify a polynomial of degree less than R with its coefficient vector in $\mathbb{F}^R$. Since $\langle h_x, f \rangle = \lambda_x f(x)$, every coefficient vector of a polynomial in $Q_E \mathbb{F}[X]_{<d}$ annihilates V_E . The MDS column property gives $\dim V_E = t$, while multiplication by the nonzero monic polynomial Q_E gives $\dim Q_E \mathbb{F}[X]_{<d} = d = R - t$. Consequently

$$V_E^\perp = \{\text{coeff}(Q_E P) : P \in \mathbb{F}[X]_{<d}\}.$$

Testing against the basis $1, X, \dots, X^{d-1}$ of $\mathbb{F}[X]_{<d}$ gives exactly the d coordinates in (DR1). $\square$

For a syndrome pair $y_0, y_1 \in \mathbb{F}^R$, and a support family $\Omega \subseteq \binom{D}{t}$, define its support-restricted finite bad-slope set by

$$(DR3) \quad Z_\Omega(y_0, y_1) = \left\{ \gamma \in \mathbb{F} : \begin{array}{l} \text{there is } E \in \Omega \text{ with } y_0 + \gamma y_1 \in V_E, \\ \{y_0, y_1\} \not\subseteq V_E \end{array} \right\}.$$

For $E \in \Omega$, abbreviate

$$u_E = \mathcal{A}_E(y_0), \quad v_E = \mathcal{A}_E(y_1).$$

Let $\mathcal{P}_j(y_1)$ be the universal pivot locus on which j is the first nonzero coordinate of v_E , and put $\Omega_j = \Omega \cap \mathcal{P}_j(y_1)$:

$$(DR4) \quad \begin{aligned} \mathcal{P}_j(y_1) &= \{E \in \binom{D}{t} : v_{E,i} = 0 \text{ for } 0 \leq i < j, \quad v_{E,j} \neq 0\}, \\ \Omega_j &= \Omega \cap \mathcal{P}_j(y_1), \end{aligned} \quad 0 \leq j < d.$$

On this chart define the *determinantal ray boundary*

$$(DR5) \quad \Delta_j(E) = (v_{E,j} u_{E,\ell} - u_{E,j} v_{E,\ell})_{\substack{0 \leq \ell < d \\ \ell \neq j}} \in \mathbb{F}^{d-1}.$$

When $d = 1$, the target in (DR5) is the trivial group and the displayed tuple is empty.

Theorem 8.2 (Exact determinantal ray compiler). *With the notation above, an error support $E \in \Omega$ supplies a finite transverse slope if and only if*

$$(DR6) \quad v_E \neq 0, \quad \text{rank} \begin{pmatrix} u_E \\ v_E \end{pmatrix} \leq 1.$$

On the pivot chart Ω_j , this is equivalent to $\Delta_j(E) = 0$, and its unique slope is $\gamma_E = -u_{E,j}/v_{E,j}$. Consequently

$$(DR7) \quad Z_\Omega(y_0, y_1) = \bigcup_{j=0}^{d-1} \left\{ -\frac{u_{E,j}}{v_{E,j}} : E \in \Omega_j, \Delta_j(E) = 0 \right\},$$

and

$$(DRC) \quad |Z_\Omega(y_0, y_1)| \leq \sum_{j=0}^{d-1} |\Omega_j \cap \Delta_j^{-1}(0)|.$$

The construction uses $d \leq n$ pivot charts and contains no unnamed finite or asymptotic multiplicity.

Proof. By Lemma 8.1, E supports the line parameter γ precisely when

$$(DR8) \quad u_E + \gamma v_E = 0.$$

If $v_E = 0$, a solution of (DR8) forces $u_E = 0$, so $y_0, y_1 \in V_E$ and the intersection is common rather than transverse. If $v_E \neq 0$, equation (DR8) has a solution if and only if u_E is a scalar multiple of v_E , which is (DR6); the scalar, and therefore the finite slope, is unique. The charts Ω_j partition the locus $v_E \neq 0$. On Ω_j , vanishing of all 2-by-2 minors using the pivot column j is equivalent to proportionality, and the proportionality factor is $-u_{E,j}/v_{E,j}$. This proves (DR7). Taking the cardinality of a union and then forgetting the slope projection gives (DRC). $\square$

Proposition 8.3 (Exact slope-incidence linearization). *With the notation of Theorem 8.2, put*

$$(DRC1) \quad \mathcal{I}_\Omega = \{(\gamma, E) \in \mathbb{F} \times \Omega : u_E + \gamma v_E = 0, v_E \neq 0\}.$$

Projection $\mathcal{I}_\Omega \rightarrow \Omega$ is injective, its slope projection is $Z_\Omega(y_0, y_1)$, and

$$(DRC2) \quad |Z_\Omega(y_0, y_1)| \leq |\mathcal{I}_\Omega| = \sum_{\gamma \in \mathbb{F}} |\{E \in \Omega : u_E + \gamma v_E = 0, v_E \neq 0\}|.$$

Thus a fixed-slope zero-fiber estimate must be aggregated over all slopes. If every fixed-slope zero fiber were bounded by $|\Omega|/q^d$, summing over the q slopes would give $|\Omega|/q^{d-1}$, not $|\Omega|/q^d$.

Proof. For $v_E \neq 0$, the vector equation determines at most one γ , proving injectivity. The exact determinantal compiler says that its slope image is precisely the transverse bad-slope set. Counting the incidence first by E and then by γ gives (DRC2); the final scale is the corresponding uniform fiber calculation. $\square$

Proposition 8.4 (The recurrence boundary is not pointwise additive). *The exact proportionality condition above does not, by itself, turn the locator-recurrence boundary into a subset-sum map. Already for $E = \{x, y\}$, one coordinate of $\mathcal{A}_E(\eta)$ is*

$$(DRC3) \quad F(x, y) = \eta_j xy - \eta_{j+1}(x + y) + \eta_{j+2}.$$

If $\eta_j \neq 0$, this function cannot be written as $c + g(x) + g(y)$ whenever the four admissible two-sets $\{x, y\}, \{x', y\}, \{x, y'\}, \{x', y'\}$ lie in the chart and $(x - x')(y - y') \neq 0$. Consequently, a proposed Fourier reduction to additive maps $E \mapsto \sum_{t \in E} g_\gamma(t)$ requires a separate coordinate theorem; it is not a consequence of slope linearization.

Proof. For a two-set, $Q_E(X) = X^2 - (x + y)X + xy$, which gives (DRC3). Its mixed difference is

$$F(x, y) - F(x', y) - F(x, y') + F(x', y') = \eta_j(x - x')(y - y').$$

Every function $c + g(x) + g(y)$ has zero mixed difference, proving the claim whenever the displayed product is nonzero. $\square$

The maps in (DR5) are explicit quadratic maps in the locator coefficients. They depend on the received syndrome pair, as they must in a worst-line theorem, but their number and degree are literal. The next corollary shows how they replace the abstract ray interface.

Theorem 8.5 (Fixed-core determinacy ray compiler). *Let $C_0 \subseteq D$ have size $c \leq k$, let $a \geq k + 1$, and fix a received line $r = (r_0, r_1)$. Suppose that for every γ in a set Z of distinct finite MCA-bad slopes we have chosen an exact-a noncommon witness $(\gamma, S_\gamma, h_\gamma)$ with $C_0 \subseteq S_\gamma$. Then*

$$(RC_{\text{core}}) \quad |Z| \leq \binom{n - c}{k - c + 1}.$$

In particular, a fixed-core cell with $k - c = o(n)$ has a direct subexponential ray payment. This statement is uniform in the received line and valid in every characteristic.

Proof. Put $d = k - c$, let $Q_{C_0}(X) = \prod_{x \in C_0} (X - x)$, and choose $g_0, g_1 \in \mathbb{F}[X]_{<k}$ interpolating r_0, r_1 , respectively, on C_0 . Write $e_i = r_i - \text{ev}_D(g_i)$. Since $h_\gamma - g_0 - \gamma g_1$ vanishes on C_0 , there is a unique $P_\gamma \in \mathbb{F}[X]_{<d}$ such that

$$h_\gamma = g_0 + \gamma g_1 + Q_{C_0} P_\gamma.$$

For $x \in D \setminus C_0$, the witness equation is

$$(*) \quad e_0(x) + \gamma e_1(x) - Q_{C_0}(x)P_\gamma(x) = 0.$$

Homogenize the parameters as $[z : \gamma : p_0 : \dots : p_{d-1}] \in \mathbb{P}^{d+1}$, and let the row indexed by $x \in D \setminus C_0$ be the linear functional

$$ze_0(x) + \gamma e_1(x) - Q_{C_0}(x) \sum_{j=0}^{d-1} p_j x^j.$$

The rows indexed by $S_\gamma \setminus C_0$ have rank exactly $d+1$. Indeed, the displayed witness parameter is a nonzero kernel vector, so the rank is at most $d+1$. If the rank were at most d , the kernel would contain an independent vector. Subtracting a multiple of the witness vector gives a nonzero kernel vector with $z = 0$. Its γ -coordinate cannot vanish: otherwise a nonzero polynomial of degree less than d would vanish at all $a - c \geq d+1$ points of $S_\gamma \setminus C_0$. Adding a nonzero multiple of this kernel vector to the witness vector therefore produces a second finite slope explained on the same support. By fixed-support slope uniqueness that support would be common, contrary to the witness hypothesis.

Choose, in a fixed order, the first $(d+1)$ -subset $T_\gamma \subseteq S_\gamma \setminus C_0$ on which the rows have rank $d+1$. Their common projective kernel is the single witness parameter, so T_γ determines γ . Hence $\gamma \mapsto T_\gamma$ is injective into $\binom{D \setminus C_0}{d+1}$, proving the bound. If $d = o(n)$, then $\log \binom{n-c}{d+1} = o(n)$. $\square$

Remark 8.6 (Cores are internal unless their census is small). The theorem pays one fixed core. It does not permit an atlas to use every positive-density subset C_0 as a separate profile: that census can be exponential. A complete atlas either routes only an algebraically specified subexponential family of cores or keeps the actual core internal to a coarse intersection-size profile.

8.2. Sharpness of support-only ray bounds.

Theorem 8.7 (Aperiodic ray obstruction below the profile frontier). *There is a sequence of smooth multiplicative rows*

$$D_n = \mathbb{B}_n^\times \subseteq \mathbb{F}_n^\times, \quad \log |\mathbb{B}_n| = O(\log n) = o(n),$$

in characteristic two, with rates tending to $1/2$, and agreements $m_n = k_n + O(n/\log n)$, for which one received line has $2^{\Omega(n)}$ distinct finite MCA-bad slopes. Every threshold- m_n witness support of that line has trivial multiplicative stabilizer, and after enlarging $\mathbb{F}_n$ with only $\log |\mathbb{F}_n| = O(n)$, the bad slopes and exact-agreement supports are in bijection. Thus smoothness, quotient-aperiodicity, Q/SP support data, and exact agreement do not force retained-support occupancy greater than one. This is a sharpness result for occupancy-based compilers, not a counterexample to a direct distinct-slope theorem at the correctly normalized profile scale.

Proof. Let $q_r = 2^r$, put $\mathbb{B}_r = \mathbb{F}_{q_r}$, $D_r = \mathbb{B}_r^\times$, and $n_r = q_r - 1$. Set

$$m_r = 2^{r-1} - 1 = \frac{n_r - 1}{2}.$$

Fix $0 < \eta < \log 2$, take $w_r = \lfloor \eta n_r / \log q_r \rfloor$, and put $k_r = m_r - w_r - 1$. Then $k_r/n_r \rightarrow 1/2$ and $m_r - k_r = w_r + 1 = O(n_r / \log n_r)$.

Pigeonholing the m_r -set locators over their $q_r^{w_r}$ prefixes gives a fiber $\mathcal{F}_r$ satisfying

$$\log |\mathcal{F}_r| \geq \log \binom{n_r}{m_r} - w_r \log q_r \geq (\log 2 - \eta)n_r - o(n_r).$$

Moreover

$$\gcd(m_r, n_r) = \gcd(2^{r-1} - 1, 2^r - 1) = 2^{\gcd(r-1, r)} - 1 = 1.$$

If an m_r -support were invariant under a nontrivial subgroup of D_r of order $c > 1$, it would be a union of c -element orbits, forcing $c \mid m_r$ and $c \mid n_r$, a contradiction. Thus every member of $\mathcal{F}_r$ is aperiodic.

Choose a finite extension $\mathbb{F}_r/\mathbb{B}_r$ with

$$|\mathbb{F}_r| > n_r + k_r \binom{|\mathcal{F}_r|}{2}.$$

Since $|\mathcal{F}_r| \leq 2^{n_r}$, the least extension degree meeting this inequality has $\log|\mathbb{F}_r| = O(n_r)$. By [Corollary 7.9](#), a separating pole gives exactly $|\mathcal{F}_r| = \exp(\Omega(n_r))$ distinct bad slopes, one for each aperiodic support, and every occupancy equals one. $\square$

Remark 8.8 (Scope of the aperiodic obstruction). The construction lies below the entropy–profile crossing: its natural prefix scale is itself exponential. It therefore does not contradict a profile-envelope upper bound proved at the correct scale. It does rule out a field-independent polynomial ray bound, division of SP by the full support slice without a retained multiplicity theorem, and the assertion that quotient-aperiodicity alone forces ray collisions.

The opposite projection extreme occurs over the coefficient field itself. It shows that a large aperiodic prefix fiber need not supply a lower reserve after first-match saturation.

Theorem 8.9 (Aperiodic one-ray saturation). *For an integer $\nu \geq 5$, put*

$$q = 2^\nu, \quad D = \mathbb{F}_q^\times, \quad n = q - 1, \quad m = 2^{\nu-1} - 1, \quad w = \lfloor n/\nu^2 \rfloor, \quad k = m - w - 1.$$

Refine the depth- w prefix by the locator constant coefficient:

$$\Psi_\nu(S) = (c_1(S), \dots, c_w(S), c_m(S)) \quad \left(S \in \binom{D}{m} \right).$$

There are $z \in \mathbb{F}_q^w$ and $c \in \mathbb{F}_q^\times$ for which

$$(SAT1) \quad \mathcal{G}_{z,c} = \{S : \Psi_\nu(S) = (z, c)\}, \quad |\mathcal{G}_{z,c}| \geq \left\lceil \binom{n}{m} q^{-(w+1)} \right\rceil = \exp((\log 2 - o(1))n).$$

Every support in $\mathcal{G}_{z,c}$ has trivial multiplicative stabilizer, but all witnesses arising from the exact prefix list project at the pole $\alpha = 0$ in the coefficient field $\mathbb{F}_q$ to the single MCA slope $-c$ for $\text{RS}_{\mathbb{F}_q}(D, k)$.

More precisely, if $\mathcal{F}_z = \Phi_w^{-1}(z)$ and $\mathcal{C}_d = \{S \in \mathcal{F}_z : c_m(S) = d\}$, then the nonempty $\mathcal{C}_d$, $d \in \mathbb{F}_q^\times$, partition the supports of all exact- m witnesses of this pole line, and the witnesses supported on $\mathcal{C}_d$ have slope image exactly $\{-d\}$. Thus at most $q - 1 = \exp(o(n))$ explicit saturation cells pay the complete line even though one cell contains exponentially many aperiodic supports.

Proof. The map Ψ_ν has at most q^{w+1} values, so pigeonholing proves the first inequality in [\(SAT1\)](#). Moreover

$$\log \binom{n}{m} = n \log 2 - O(\log n), \quad (w+1) \log q \leq (n/\nu + \nu) \log 2 = o(n),$$

which proves its asymptotic form. Also

$$\gcd(m, n) = \gcd(2^{\nu-1} - 1, 2^\nu - 1) = 2^{\gcd(\nu-1, \nu)} - 1 = 1.$$

An invariant support would be a union of orbits of a nontrivial subgroup whose order divides both m and n , so every m -support is multiplicatively aperiodic.

Let $U_z = X^m + \sum_{i=1}^w z_i X^{m-i}$ and $P_S = U_z - Q_S$. Prefix cancellation gives $\deg P_S \leq k$, and $U_z - P_S = Q_S$ shows that P_S agrees with U_z exactly on S . At the only pole of the coefficient field outside D , namely zero,

$$P_S(0) = U_z(0) - Q_S(0) = -c_m(S).$$

The pole-line calculation therefore maps every $\mathcal{C}_d$ to $-d$. Conversely, an exact- m explanation on this line gives a polynomial $P = Xh + \gamma$ of degree at most k agreeing with U_z on m points. The exact prefix-list bijection makes it the unique $P_S = U_z - Q_S$ for some $S \in \mathcal{F}_z$, and then $\gamma = P(0) = -c_m(S)$. This proves coverage and the asserted one-slope images. $\square$

Remark 8.10 (Two incompatible aperiodic inferences). [Theorem 8.7](#) makes a separating extension pole inject an exponentially large aperiodic fiber into distinct slopes, while [Theorem 8.9](#) makes a coefficient-field pole collapse such a refined fiber to one slope. Aperiodicity alone therefore implies neither ray expansion nor ray collision. A lower compiler needs a separating-pole estimate or a quantitative exclusion of earlier saturation cells; an upper compiler needs an actual determinant or ray-image bound.

8.3. Smooth and circle evaluation domains. This subsection records the domain geometry behind the exact quotient constructions. The multiplicative and circle cases are treated simultaneously through uniform-fiber folding maps. The point is not merely that such maps exist, but that their fibers are complete inside the evaluation domain and their locators have a *lacunary* top part, meaning that prescribed gaps occur among the highest-degree coefficients. These two facts make quotient families visible in the witness moduli and keep their locator prefixes stable under planted cores.

Definition 8.11 (Complete-fiber folding map and smooth multiplicative coset). *Let $D \subseteq \mathbb{B}$ be finite. A polynomial map $\varphi : D \rightarrow Q := \varphi(D)$ is a c -fold complete-fiber folding map if every $y \in Q$ has exactly c preimages in D . A support is complete at this folding scale if it is a union of whole fibers, equivalently if it is $\varphi^{-1}(E)$ for some $E \subseteq Q$. Passing from D to Q , and from a complete support to E , is the corresponding quotient or folding.*

A multiplicative coset is a set $D = \theta H \subseteq \mathbb{B}^\times$, where $H \leq \mathbb{B}^\times$ is a subgroup and $\theta \in \mathbb{B}^\times$. For every divisor $c \mid |H|$, the restriction

$$\pi_c : D \longrightarrow D^{(c)}, \quad \pi_c(x) = x^c, \quad D^{(c)} = \{x^c : x \in D\},$$

is a c -fold complete-fiber folding map. We call D smooth at scale c when this power folding is among the quotients retained by the row, and smooth for a family of scales when it is smooth at every scale in that family. Thus the adjective smooth here is an explicit uniform-fiber property, not a claim about nonsingularity of an algebraic variety. The locator of a complete support is $V_E(X^c)$; its nonleading terms occur only in degree gaps divisible by c . This controlled pattern of missing top coefficients is what lacunary means in this section.

Definition 8.12 (Circle domain and circle code). *Let $\mathbb{B}_0$ have odd characteristic and let $\mathcal{C}/\mathbb{B}_0$ be the affine conic $\mathcal{C} : x^2 + y^2 = 1$. A standard circle domain is a finite set $E \subseteq \mathcal{C}(\mathbb{B}_0)$. If $\mathbb{F} \supseteq \mathbb{B}_0$, the degree- w circle code on E is*

$$\mathcal{C}_w(\mathbb{F}, E) = \{(f(P))_{P \in E} : f \in \text{im}(\mathbb{F}[x, y]_{\leq w} \rightarrow \mathbb{F}[x, y]/(x^2 + y^2 - 1))\}.$$

The degree is the total-degree filtration before passage to the quotient ring. A circle x -domain is instead a set of first coordinates obtained from the torus construction below; an RS code on that x -domain and a bivariate circle code are different presentations until a diagonal equivalence such as [Lemma 8.18](#) has been exhibited.

Definition 8.13 (Norm-one torus, twin coset, and Chebyshev domain). *Following the circle-domain geometry of [\[HLP24\]](#), let $p \equiv 3 \pmod{4}$, choose $i \in \mathbb{F}_{p^2}$ with $i^2 = -1$, and write*

$$\mathbb{U} = \ker\left(N_{\mathbb{F}_{p^2}/\mathbb{F}_p} : \mathbb{F}_{p^2}^\times \rightarrow \mathbb{F}_p^\times\right) = \{u \in \mathbb{F}_{p^2}^\times : u^{p+1} = 1\}.$$

The algebraic kernel of the norm morphism is the norm-one torus; $\mathbb{U}$ is its cyclic group of $\mathbb{F}_p$ -rational points. Put $\chi(u) = (u + u^{-1})/2 \in \mathbb{F}_p$. If $H \leq \mathbb{U}$ and $g \in \mathbb{U}$ satisfies $g^2 \notin H$, then

$$\mathcal{D}(g, H) = gH \sqcup g^{-1}H$$

is a disjoint union exchanged by $u \mapsto u^{-1}$; it is called a twin coset. Its projected set

$$D(g, H) = \chi(\mathcal{D}(g, H)) \subseteq \mathbb{F}_p$$

has cardinality $|H|$ and is the Chebyshev twin-coset x -domain.

For $c \geq 1$, let T_c be the Chebyshev polynomial normalized by

$$T_c\left(\frac{u + u^{-1}}{2}\right) = \frac{u^c + u^{-c}}{2}.$$

Thus $T_c \circ \chi = \chi \circ (u \mapsto u^c)$. Whenever $c \mid |H|$ and the two image cosets remain disjoint, the restriction of T_c to $D(g, H)$ is the Chebyshev folding map; a Chebyshev twin-coset domain equipped with these complete-fiber maps is the circle analogue of a smooth multiplicative coset.

Definition 8.14 (Map-smooth domain). *Let $\mathbb{B} \subseteq \mathbb{F}$ be finite fields, let $\varphi \in \mathbb{B}[X]$ have degree $c \geq 1$, and let $D \subseteq \mathbb{B}$ have size n . The set D is (φ, c) -smooth if every nonempty fiber $\varphi^{-1}(y) \cap D$, $y \in \varphi(D)$, has exactly c elements. We write $Q = \varphi(D)$ and $N = n/c$.*

A multiplicative coset $D = \theta H \subseteq \mathbb{B}^\times$ is (X^a, a) -smooth for every divisor $a \mid |H|$. The circle case uses Chebyshev maps rather than power maps; see [Lemma 8.17](#). The following lemma is the common quotient-support construction. It is included because it produces explicit large lists and because the same lacunarity controls their locator prefixes.

Lemma 8.15 (Map-smooth fiber construction). *Let D be (φ, a) -smooth, $|D| = n$, and $C^+ = \text{RS}_{\mathbb{F}}(D, k+1)$. Put $\ell = \lfloor k/a \rfloor + 2$ and $A = a\ell$. If $\ell \leq N-1$, then there is $z \in \mathbb{B}$ and a word $u_z \in \mathbb{B}^D$ such that u_z has at least $\binom{N}{\ell}/|\mathbb{B}|$ distinct codewords of C^+ agreeing with it on at least A positions. Moreover $k+a+1 \leq A \leq k+2a$, with equality $A = k+2a$ if $a \mid k$.*

Proof. For each ℓ -subset $E \subseteq Q$, let $P_E(Y) = \prod_{b \in E} (Y - b) = Y^\ell - e_1(E)Y^{\ell-1} + R_E(Y)$, with $\deg R_E \leq \ell - 2$. The composed locator $P_E(\varphi(X))$ vanishes on the union of the a -point fibers above E , hence on $A = a\ell$ points of D . Set $z_E = -e_1(E)$ and $c_E = -R_E(\varphi)$. Since $\deg c_E \leq a(\ell - 2) \leq k$, the word c_E lies in C^+ , and it agrees on that fiber union with $u_{z_E}(x) = \varphi(x)^\ell + z_E \varphi(x)^{\ell-1}$. Among the $\binom{N}{\ell}$ subsets E , some value of $z_E \in \mathbb{B}$ occurs at least $\binom{N}{\ell}/|\mathbb{B}|$ times. If two such subsets with the same z gave the same codeword, then $R_E(\varphi) = R_{E'}(\varphi)$, hence $R_E = R_{E'}$ because composition with a nonconstant polynomial is injective, and then $P_E = P_{E'}$. Thus the codewords are distinct. The bounds on A follow from $\ell = \lfloor k/a \rfloor + 2$. $\square$

Proposition 8.16 (Collision-aware lower cap for map-smooth domains). *Let D be (φ, a) -smooth, $n < |\mathbb{F}| = q$, and let $C = \text{RS}_{\mathbb{F}}(D, k)$. With ℓ, A as in [Lemma 8.15](#), assume $\ell \leq N-1$, and put*

$$L = \left\lceil \binom{N}{\ell} |\mathbb{B}|^{-1} \right\rceil.$$

Then for every integer threshold $k+1 \leq m \leq A$,

$$B_C^{\text{MCA}}(m) \geq \left\lceil \frac{L(q-n)}{q-n+k(L-1)} \right\rceil.$$

Proof. The list in [Lemma 8.15](#) has size at least L at agreement A . Apply the collision-aware pole conversion [Theorem 7.2](#). Its line pair is not simultaneously explained on any $k+1$ points, so an agreement set of size A also witnesses badness for every threshold m in the displayed range. $\square$

We now verify the complete-fiber property for the construction in [Definition 8.13](#). Thus $\mathcal{D} = gH \sqcup g^{-1}H \subseteq \mathbb{U}$, $D = \chi(\mathcal{D})$, and $T_a(\chi(u)) = \chi(u^a)$.

Lemma 8.17 (Chebyshev smoothness). *Let $\mathcal{D} = gH \cup g^{-1}H \subseteq \mathbb{U}$ be a twin coset, let $D = \chi(\mathcal{D})$, and let $a \mid |H|$. If $g^{2a} \notin H^{(a)} = \{h^a : h \in H\}$, then D is (T_a, a) -smooth. If $\text{ord}(g) \nmid 2|H|$, this holds for all $a \mid |H|$.*

Proof. The map $u \mapsto u^a$ has kernel of size a on $\mathbb{U}$, and that kernel lies in H . Hence it maps gH and $g^{-1}H$ onto $g^a H^{(a)}$ and $g^{-a} H^{(a)}$, respectively, with fibers of size a . The hypothesis says these image cosets are disjoint, so the image is again a twin coset. Since χ is exactly two-to-one on both twin cosets and identifies only inverse points, the equation $T_a(x) = y$ on D has exactly a solutions for each $y \in T_a(D)$. Taking $a = |H|$ shows that the all-scale condition is equivalent to $g^{2|H|} \neq 1$, and the implication to every smaller divisor follows by raising to $|H|/a$. $\square$

Lemma 8.18 (Diagonal uniformization of circle codes). *Assume $\mathbb{F} \supseteq \mathbb{F}_{p^2}$. Let $E \subseteq \{(x, y) \in \mathbb{F}_p^2 : x^2 + y^2 = 1\}$, define $\iota(x, y) = x + iy$, put $E' = \iota(E) \subseteq \mathbb{U}$, and let $\mathcal{C}_w(\mathbb{F}, E)$ be the circle code of [Definition 8.12](#). Under the coordinate bijection ι and the diagonal multiplier $t_u = u^{-w}$,*

the code $\mathcal{C}_w(\mathbb{F}, E)$ is $t \circ \text{RS}_{\mathbb{F}}(E', 2w + 1)$. Consequently list sizes, ordinary correlated-agreement (CA) numerators, and mutual-correlated-agreement (MCA) numerators are preserved.

Proof. The map $u = x + iy$ identifies $\mathbb{F}[x, y]/(x^2 + y^2 - 1)$ with $\mathbb{F}[u, u^{-1}]$. The subspace of total degree at most w maps to $u^{-w}\mathbb{F}[u]_{\leq 2w}$, and both spaces have dimension $2w + 1$. Multiplication by the nonzero diagonal vector $(u^{-w})_{u \in E'}$ preserves coordinatewise equality and hence every agreement notion used here. Therefore it preserves list, CA, and MCA counts exactly. $\square$

The preceding uniformization is convenient when i belongs to the challenge field. The following form works over every odd-characteristic field and will also be used for scalar extensions of odd degree.

Lemma 8.19 (Invariance under monomial equivalence). *Let $C \subseteq \mathbb{F}^D$, let $\pi : D' \rightarrow D$ be a bijection, and let $(d_x)_{x \in D'} \in (\mathbb{F}^\times)^{D'}$. Define*

$$C' = \{(d_x c_{\pi(x)})_{x \in D'} : c \in C\}.$$

The same coordinate permutation and diagonal multiplication give a bijection between received pairs for C and C' , preserving the bad-slope set on every support and for every challenge set. In particular $B_{C, \Gamma}^{\text{MCA}}(a) = B_{C', \Gamma}^{\text{MCA}}(a)$.

Proof. For every word u , codeword c , and support S , equality $u|_S = c|_S$ is equivalent after the displayed coordinate change to equality on $\pi^{-1}(S)$. The transformation is $\mathbb{F}$ -linear, so it commutes with $r_0 + \gamma r_1$. It therefore preserves both the single-line-point explanation and simultaneous explanation of the pair on the corresponding support, slope by slope. $\square$

Corollary 8.20 (Generalized Reed–Solomon rows). *Every generalized Reed–Solomon code obtained from $\text{RS}_{\mathbb{F}}(D, k)$ by a coordinate permutation and nonzero column multipliers has exactly the same CA and MCA numerators as that RS code. In particular all exact staircases and target compilers in this paper transfer verbatim.*

Proof. This is Lemma 8.19 applied to the defining monomial equivalence. $\square$

Proposition 8.21 (Stereographic circle equivalence). *Let $\mathbb{F}$ have odd characteristic, let $E \subseteq \{(x, y) \in \mathbb{F}^2 : x^2 + y^2 = 1\}$ avoid $(-1, 0)$, and put*

$$t = \frac{y}{1+x}, \quad T = t(E).$$

If $2w + 1 \leq |E|$, then

$$(SC1) \quad \mathcal{C}_w(\mathbb{F}, E) = \text{diag}((1+t^2)^{-w})_{t \in T} \text{RS}_{\mathbb{F}}(T, 2w + 1)$$

after the coordinate bijection $E \rightarrow T$. Hence every list, CA, and support-wise MCA bad-slope set is preserved exactly.

Proof. The inverse parametrization is

$$x = \frac{1-t^2}{1+t^2}, \quad y = \frac{2t}{1+t^2},$$

and $1+t^2 = 2/(1+x) \neq 0$ on E . Modulo the circle equation, the degree-at-most- w filtered space is

$$\{f_0(x) + yf_1(x) : \deg f_0 \leq w, \deg f_1 \leq w-1\},$$

with dimension $2w + 1$. After substitution and multiplication by $(1+t^2)^w$, every such function becomes a polynomial of degree at most $2w$. The substitution is injective in the localized coordinate ring and the dimensions agree, so all of $\mathbb{F}[t]_{\leq 2w}$ occurs. Equation (SC1) follows, and Lemma 8.19 proves the last assertion. $\square$

Corollary 8.22 (Exact circle staircase). *Let E, w be as in Proposition 8.21, put $n = |E|$ and $k = 2w + 1 < n$. For every nonempty $\Gamma \subseteq \mathbb{F}$ and every $0 \leq r \leq n - k - 1$,*

$$(SC2) \quad (n-r)^2 \geq n(k+r) \implies B_{\mathcal{C}_w(\mathbb{F}, E), \Gamma}^{\text{MCA}}(n-r) = \min\{|\Gamma|, r+1\}.$$

The same equality holds under the hypotheses of Theorem 5.2. Consequently standard circle codes admitting such a projection chart have the positive-relative-radius staircase of Corollary 5.4.

Proof. Apply [Lemma 8.19](#) and then [Theorems 4.6](#) and [5.2](#) to the RS presentation in [\(SC1\)](#). $\square$

For a set $\mathcal{P}$ of primes, its *relative prime density* is

$$\text{dens}_{\text{pr}}(\mathcal{P}) = \lim_{X \rightarrow \infty} \frac{\#\{p \leq X : p \in \mathcal{P}\}}{\pi(X)},$$

when the limit exists. The prime number theorem in arithmetic progressions gives density $1/\varphi(M)$ to every reduced residue class modulo M [\[Dav00\]](#). We write $\overline{\text{dens}}_{\text{pr}}$ for the same expression with limsup in place of the limit.

Theorem 8.23 (Fixed-length positive-density prime families). *Fix $n = 2^m$, $m \geq 2$.*

- (a) *The primes $p \equiv 1 \pmod{n}$, of relative prime density $1/\varphi(n) = 2/n$, support the unique subgroup $H \leq \mathbb{F}_p^\times$ of order n . Every exact RS conclusion of [Theorems 4.6](#) and [5.2](#) holds on H and on each of its multiplicative cosets.*
- (b) *The primes $p \equiv -1 \pmod{2n}$, of density $1/\varphi(2n) = 1/n$, support a standard circle coset E of size n . For every odd $k = 2w + 1 < n$, the circle code $\mathcal{C}_w(\mathbb{F}_{p^2}, E)$ has both exact staircases.*
- (c) *The primes $p \equiv -1 \pmod{4n}$, of density $1/\varphi(4n) = 1/(2n)$, support a Chebyshev-smooth circle x -domain $D \subseteq \mathbb{F}_p$ of size n , and $\text{RS}_{\mathbb{F}_p}(D, k)$ has both exact staircases for every $k < n$.*

Proof. Part (a) follows from cyclicity of $\mathbb{F}_p^\times$. For part (b), let $\mathbb{U} \leq \mathbb{F}_{p^2}^\times$ be the norm-one group, of order $p + 1$, choose $g \in \mathbb{U}$ of order $2n$, and put $H = \langle g^2 \rangle$. The coset $E' = gH$ has size n , is stable under inversion, and avoids -1 . Its inverse image under $u = x + iy$ consists of n points of the circle over $\mathbb{F}_p$. [Lemma 8.18](#) identifies its circle code with a diagonal translate of $\text{RS}_{\mathbb{F}_{p^2}}(E', 2w + 1)$.

For part (c), choose $g \in \mathbb{U}$ of order $4n$, put $H = \langle g^4 \rangle$, and use the twin coset $gH \sqcup g^{-1}H$. Its χ -image has size n , and $\text{ord}(g) = 4n \nmid 2n$, so [Lemma 8.17](#) proves smoothness at every dyadic scale. The density assertions are the prime number theorem in the three displayed reduced residue classes. $\square$

Corollary 8.24 (Finite smooth and circle families beyond the prior window). *Let $n = 32, k = 15, r = 6$. For every prime $p \equiv 1 \pmod{32}$, let $H \leq \mathbb{F}_p^\times$ have order 32 and put $C_p^+ = \text{RS}_{\mathbb{F}_p}(H, 15)$. For every prime $p \equiv -1 \pmod{64}$, construct the circle coset E of [Theorem 8.23\(b\)](#) and put $C_p^- = \mathcal{C}_7(\mathbb{F}_{p^2}, E)$. Then*

$$(SC3) \quad B_{C_p^+}^{\text{MCA}}(26) = B_{C_p^-}^{\text{MCA}}(26) = 7.$$

At targets $6/p$ and $6/p^2$, respectively, both complete safe sets are $[0, 3/16)$, with unsafe endpoint. The two supporting prime sets have relative densities $1/16$ and $1/32$.

Proof. Here $R = 17$ and

$$(n - r)^2 = 26^2 = 676 > 672 = 32(15 + 6).$$

Thus [Theorem 4.6](#) and [Corollary 8.22](#) give the numerator seven for the smooth and circle rows. Radius five is in the deep range and has numerator six; the tangent floor at radius six and monotonicity give the stated safe sets. This example lies beyond the elementary one-third range because $3r = 18 > R$. It is not certified by the specialized BCHKS estimate [\(LC5\)](#), whose first term is $176/15 > 7$. The respective densities are $1/\varphi(32) = 1/16$ and $1/\varphi(64) = 1/32$. $\square$

Corollary 8.25 (Concrete Mersenne-31 circle rows). *Let $p = 2^{31} - 1$, the Mersenne prime used in [\[HLP24\]](#), and let $\mathbb{U}$ be its norm-one circle group. Choose a generator g , put $H_0 = \langle g^2 \rangle$, and let $E \subseteq \{(x, y) \in \mathbb{F}_p^2 : x^2 + y^2 = 1\}$ be the circle domain corresponding to gH_0 , so $|E| = 2^{30}$. This domain avoids $(-1, 0)$. Over $\mathbb{F}_{p^5}$, the circle code with $w = 2^{28}$ and $k = 2^{29} + 1$ has, at target 2^{-128} , the exact safe set*

$$(SC4) \quad \left[0, \frac{1}{8} - 2^{-30}\right).$$

Moreover, if $H = \langle g^4 \rangle$, the Chebyshev twin-coset $gH \sqcup g^{-1}H = gH_0$ has x -domain $D = \chi(gH_0)$ of size 2^{29} , and $\text{RS}_{\mathbb{F}_{p^5}}(D, 2^{27})$ has exact safe set

$$(SC5) \quad \left[0, \frac{1}{4} - 2^{-29}\right).$$

The first code has rate slightly above $1/2$; the second has exact rate $1/4$ but is a circle-smooth x -domain rather than a literal multiplicative-domain Prize row.

Proof. The group $\mathbb{U}$ has order $p+1 = 2^{31}$. Since $-1 = g^{2^{30}} \in H_0$, neither 1 nor -1 belongs to the disjoint coset gH_0 ; in particular E avoids the pole $(-1, 0)$ of the stereographic chart. Also $gH_0 = gH \sqcup g^{-1}H$. Inversion acts freely on this coset and $\chi(u) = (u + u^{-1})/2$ identifies precisely the pairs $\{u, u^{-1}\}$, proving $|D| = 2^{29}$. The relation $T_a \circ \chi = \chi \circ (u \mapsto u^a)$ and Lemma 8.17 give the asserted dyadic smoothness.

Binomial expansion gives $B = \lfloor p^5/2^{128} \rfloor = 2^{27} - 1$. At radius $B - 1$, the quadratic margin $(n - r)^2 - n(k + r)$ for the bivariate row is $162129591417176068 > 0$, so Corollary 8.22 and the tangent floor give (SC4). Here $p^5 \equiv 3 \pmod{4}$, so $i \notin \mathbb{F}_{p^5}$; the applicable equivalence is the stereographic one in Proposition 8.21, not torus uniformization. For the x -domain row the same budget lies inside the quadratic staircase at rate $1/4$, since its margin at radius $B - 1$ is $18014401193836548 > 0$. The identical target argument gives (SC5). $\square$

Corollary 8.26 (Every odd-characteristic extension-field tower). *Fix an odd prime p , put $n_e = 2^e$ and $q_e = p^{n_e}$, and assume $e \geq 7$. Then $4n_e \mid q_e - 1$. Over $\mathbb{F}_{q_e}$ there exist both a multiplicative subgroup RS row of length n_e and a Chebyshev-smooth circle x -domain row of that length. At every official rate and every $0 \leq r \leq r_p^\#$, both satisfy*

$$(SC6) \quad e_{\text{MCA}}(C_e, r/n_e) = \frac{r+1}{q_e}.$$

Thus the construction exists in every odd characteristic and beats every fixed error target throughout every compact subinterval below half the minimum distance for large e .

Proof. The lifting-the-exponent identity gives

$$v_2(p^{2^e} - 1) = v_2(p - 1) + v_2(p + 1) + e - 1 \geq e + 2.$$

Cyclicity produces the multiplicative subgroup and an element of order $4n_e$. Since n_e is even, $q_e \equiv 1 \pmod{4}$; choose $i \in \mathbb{F}_{q_e}$ with $i^2 = -1$. The map

$$u \mapsto \left(\frac{u + u^{-1}}{2}, \frac{u - u^{-1}}{2i} \right)$$

identifies $\mathbb{F}_{q_e}^\times$ with the split circle over $\mathbb{F}_{q_e}$. If g has order $4n_e$ and $H = \langle g^4 \rangle$, the twin coset $gH \sqcup g^{-1}H$ projects to an n_e -point x -domain. The proof of Lemma 8.17, which uses only cyclicity and $T_a \circ \chi = \chi \circ (u \mapsto u^a)$, applies verbatim and proves Chebyshev smoothness at every dyadic scale. Apply Corollary 5.5. Since $q_e = p^{n_e}$, the normalized error tends to zero exponentially. $\square$

Proposition 8.27 (No growing positive-density prime-field family). *Let $\mathcal{P}$ be a set of primes and assign a dyadic integer $n_p \rightarrow \infty$ along $\mathcal{P}$. If $n_p \mid p - 1$ for every $p \in \mathcal{P}$, or $n_p \mid p + 1$ for every $p \in \mathcal{P}$, then $\mathcal{P}$ has upper relative prime density zero. If in addition p/n_p is bounded, there are only $O(\log X)$ possible primes of the indicated form below X .*

Proof. For every fixed E , all sufficiently large assigned lengths are divisible by 2^E . Hence the primes eventually lie in the single class $1 \pmod{2^E}$, or the single class $-1 \pmod{2^E}$, whose relative prime density is $1/\varphi(2^E) = 2^{1-E}$. Letting $E \rightarrow \infty$ proves the first assertion. Under the bounded-ratio hypothesis, $p \mp 1 = a2^e$ with a in a fixed finite set; there are only $O(\log X)$ choices of (a, e) below X . $\square$

Remark 8.28 (Prize scope). The prime-density statements fix n ; their densities tend to zero as n grows. The extension towers have exponential field size and soon leave the $q < 2^{256}$ Prize box.

Finally the target $6/p^2$ in [Corollary 8.24](#) varies with p . None of these facts may be rephrased as a positive-density growing prime-field solution at the fixed target 2^{-128} .

8.4. Quotient–remainder profiles and partial occupancy. The quotient and remainder variables can be separated exactly at the locator-prefix level. This removes a spurious factor coming from summing over marked remainder sets, but it also exposes a genuine scaled quotient-coefficient-field term which prevents an unrestricted comparison with the identity profile.

Definition 8.29 (Quotient/remainder profile and field drop). *Let $\phi : D \rightarrow Q$ be a c -fold complete-fiber folding map in the sense of [Definition 8.11](#). A complete-fiber-plus-remainder support is*

$$S = \phi^{-1}(E) \sqcup R,$$

where $E \subseteq Q$ and $R \subseteq D \setminus \phi^{-1}(E)$. The set R is the support remainder: it records the selected points in fibers that are not declared complete. A quotient/remainder profile fixes ϕ , its degree c , the size $r = |R|$, the relevant coefficient field, and any planted remainder data, while E and the unfixed part of R range subject to the displayed disjointness. A fixed- R profile is the subprofile in which the remainder support itself is part of the label. The adjective Euclidean used below refers to separating the locator factor of degree $r < c$ from the composed quotient locator; it does not mean that an arbitrary support remainder has size less than c .

We call $\mathbb{B}_\phi \subseteq \mathbb{B}$ a scaled quotient coefficient field when, for some $\eta \in \mathbb{B}^\times$,

$$Q = \phi(D) \subseteq \eta \mathbb{B}_\phi.$$

When this field is recorded in a profile, it is taken minimal among the subfields with this property. Then the j -th elementary quotient coefficient lies in $\eta^j \mathbb{B}_\phi$. A field drop is the use of a proper such subfield. It is a positive-rate field drop along a sequence when

$$1 - \frac{\log |\mathbb{B}_\phi|}{\log |\mathbb{B}|}$$

stays bounded away from zero and a positive-density number on the entropy scale of quotient-prefix slots is live. A field drop changes the number of available coefficient values and is therefore recorded explicitly.

Definition 8.30 (Balanced quotient core). A split locator is a monic squarefree polynomial all of whose roots lie in the evaluation domain D . After common locator factors, fixed quotient data, and planted remainder points have been removed, let A and B be two monic split locators of the same degree e . They form a depth- w balanced core if they have the same first w nonleading coefficients, equivalently

$$\deg(A - B) \leq e - w - 1.$$

It is a balanced quotient core when the remaining locator factors are pullbacks through the same folding map ϕ . The word balanced records equality of degree and leading normalization; the core consists of the roots still allowed to move after common and planted factors are removed. If the resulting projective locator family is one-dimensional it is a split pencil $Q_0 + \lambda Q_1$; in higher dimension this definition by itself supplies no distinct-ray bound. The exact determinant-pivot compiler still applies, but its boundary maps require the analytic payment stated in (A6).

For a monic polynomial

$$A(X) = X^d + a_1 X^{d-1} + \cdots + a_d$$

and an integer $0 \leq u \leq d$, write $\text{pref}_u(A) = (a_1, \dots, a_u)$. Equivalently, if $A^\vee(Z) := Z^d A(Z^{-1})$, then $\text{pref}_u(A)$ is the coefficient vector of $A^\vee(Z) \bmod Z^{u+1}$, with its constant coefficient omitted.

Theorem 8.31 (Exact quotient–remainder prefix normal form). *Let $\mathbb{B} \subseteq \mathbb{F}$ be finite fields, let $D \subseteq \mathbb{B}$ have n distinct elements, and let $\phi \in \mathbb{B}[X]$ be monic of degree $c \geq 2$. Assume that the restriction*

$$\phi : D \longrightarrow Q := \phi(D)$$

has complete fibers of size c ; in particular, $N := |Q| = n/c$. Fix integers m, r with $0 \leq r < c$ and $cm + r \leq n$. For

$$E \in \binom{Q}{m}, \quad R \in \binom{D \setminus \phi^{-1}(E)}{r},$$

put

$$\begin{aligned} V_E(Y) &:= \prod_{y \in E} (Y - y) = Y^m + \sum_{j=1}^m v_j(E) Y^{m-j}, \\ P_R(X) &:= \prod_{x \in R} (X - x) = X^r + \sum_{i=1}^r p_i(R) X^{r-i}, \\ L_{E,R}(X) &:= P_R(X) V_E(\phi(X)). \end{aligned} \tag{QR1}$$

Then $L_{E,R}$ is the monic locator of the support $\phi^{-1}(E) \sqcup R$, of degree $A = cm + r$.

Let $0 \leq w \leq A$ and put $d = \lfloor w/c \rfloor$.

- (i) If $w \geq r$, every nonempty fiber of $(E, R) \mapsto \text{pref}_w(L_{E,R})$ determines R uniquely and, after R is recovered, is exactly one quotient-prefix fiber. More precisely, for every prefix value z there are unique data R_z and $\eta_z \in \mathbb{B}^d$, whenever the fiber is nonempty, such that

$$\begin{aligned} \{(E, R) : \text{pref}_w(L_{E,R}) = z\} \\ = \{(E, R_z) : E \in \binom{Q \setminus \phi(R_z)}{m}, \text{pref}_d(V_E) = \eta_z\}. \end{aligned} \tag{QR2}$$

Thus no factor counting possible remainder sets occurs in a fixed global prefix fiber.

- (ii) If $w < r$, then necessarily $w < c$, and the quotient set E is invisible to the first w coefficients. There is a fixed unitriangular bijection $T_{\phi,m,w} : \mathbb{B}^w \rightarrow \mathbb{B}^w$ such that

$$\text{pref}_w(L_{E,R}) = T_{\phi,m,w}(\text{pref}_w(P_R)) \tag{QR3}$$

for every admissible (E, R) . Consequently, with the convention $\binom{u}{v} = 0$ for $v > u$,

$$\begin{aligned} \#\{(E, R) : \text{pref}_w(L_{E,R}) = z\} \\ = \sum_{\substack{R \in \binom{P}{r} \\ \text{pref}_w(P_R) = T_{\phi,m,w}^{-1}(z)}} \binom{N - |\phi(R)|}{m}. \end{aligned} \tag{QR4}$$

Thus this case is a remainder-prefix problem, multiplied by the number of quotient sets still available after R is fixed.

Proof. For a monic degree- e polynomial A , set $A^\vee(Z) = Z^e A(Z^{-1})$. Also set

$$\Phi(Z) := Z^c \phi(Z^{-1}) = 1 + \phi_1 Z + \cdots + \phi_c Z^c.$$

Taking reciprocal polynomials in (QR1) gives the exact formal-power-series identity

$$L_{E,R}^\vee(Z) = P_R^\vee(Z) \left(\Phi(Z)^m + \sum_{j=1}^m v_j(E) Z^{cj} \Phi(Z)^{m-j} \right). \tag{QR5}$$

All factors on the right have constant coefficient one except for the displayed scalar coefficients $v_j(E)$.

Suppose first that $w \geq r$. Since $r < c$, no term containing a v_j occurs below degree c . For $1 \leq i \leq r$, the coefficient of Z^i on the right of (QR5) is $p_i(R)$ plus an expression depending only on $p_1(R), \dots, p_{i-1}(R)$, on m , and on Φ . The first r coefficients therefore recover $p_1(R), \dots, p_r(R)$ successively. They recover the monic polynomial P_R , hence its root set R , uniquely.

We may now divide $L_{E,R}^\vee$ by the unit $P_R^\vee$ modulo Z^{w+1} . In the remaining factor of (QR5), the coefficient of Z^{cj} is $v_j(E)$ plus an expression depending only on $v_1(E), \dots, v_{j-1}(E)$, on m , and on Φ . Hence the coefficients at degrees $c, 2c, \dots, dc$ recover $v_1(E), \dots, v_d(E)$ successively. Conversely, P_R and these d coefficients determine the right side of (QR5) modulo Z^{w+1} , because every term containing v_j with $j > d$ starts in degree $cj > w$. This proves (QR2); admissibility of E is exactly the condition $E \cap \phi(R) = \emptyset$.

If $w < r < c$, every term involving a v_j starts in degree at least $c > w$, so modulo Z^{w+1} the bracket in (QR5) is the fixed unit $\Phi(Z)^m$. Multiplication by this unit sends the first w coefficients of $P_R^\vee$ to the first w coefficients of $L_{E,R}^\vee$ by a unitriangular transformation. It is independent of E and invertible. For a fixed R , the admissible E are precisely the m -subsets of $Q \setminus \phi(R)$, of which there are $\binom{N-|\phi(R)|}{m}$. Summing over the indicated remainder-prefix fiber proves (QR4). $\square$

Proposition 8.32 (Complete support factorization and its limitation). *Under the complete-fiber hypotheses of Theorem 8.31, every support $S \subseteq D$ has the unique decomposition*

$$(QR5a) \quad E(S) = \{y \in Q : \phi^{-1}(y) \subseteq S\}, \quad R(S) = S \setminus \phi^{-1}(E(S)),$$

and $Q_S(X) = P_{R(S)}(X)V_{E(S)}(\phi(X))$. The remainder meets every nonfull ϕ -fiber in at most $c-1$ points, but its total size may be $\Theta(n)$. The reciprocal identity (QR5) remains valid for this arbitrary remainder. When $|R(S)| \geq c$, however, the quotient coefficients and the remainder coefficients begin to interlace at degree c , so the triangular recovery theorem above does not give a comparison without an additional partial-occupancy atlas.

Proof. A fiber is put into $E(S)$ exactly when all of its points lie in S ; all other selected points form $R(S)$. This proves existence, disjointness, the occupancy bound, and uniqueness. Multiplying the corresponding linear factors gives the locator identity. The reciprocal calculation used for (QR5) is purely multiplicative and does not require $\deg P_R < c$. If $\deg P_R \geq c$, both $p_c(R)$ and $v_1(E)$ occur in degree c , which proves the stated limitation of the triangular separation. $\square$

Theorem 8.33 (Exact partial-occupancy disintegration). *Let $D = D_0 \sqcup X$, where $|X| = b$, and let $\phi : D_0 \rightarrow Q$ have exactly $N = |Q|$ fibers, each of cardinality c . For $S \subseteq D$, put*

$$X_S = S \cap X, \quad E(S) = \{y \in Q : \phi^{-1}(y) \subseteq S\}, \quad R(S) = (S \cap D_0) \setminus \phi^{-1}(E(S)).$$

Write

$$t = |X_S|, \quad m = |E(S)|, \quad p = |\phi(R(S))|, \quad r = |R(S)|.$$

Then this datum is unique, every fiber meeting $R(S)$ contains between one and $c-1$ points of $R(S)$, and $|S| = t + cm + r$. If $\Omega_{t,m,p,r}$ denotes the family of supports with these four parameters, then

$$(PO1) \quad |\Omega_{t,m,p,r}| = \binom{b}{t} \binom{N}{p} \binom{N-p}{m} [x^r]((1+x)^c - 1 - x^c)^p.$$

Consequently the nonempty families $\Omega_{t,m,p,r}$ partition $\binom{D}{a}$, and their exact add-back is

$$(PO2) \quad \sum_{\substack{t,m,p,r \\ t+cm+r=a}} |\Omega_{t,m,p,r}| = \binom{cN+b}{a}.$$

If ϕ is the restriction of a monic degree- c polynomial and every fiber $\phi^{-1}(y)$ is the complete simple root set in D_0 of $\phi(X) - y$, then their locators satisfy

$$Q_S = Q_{X_S} P_{R(S)} V_{E(S)}(\phi).$$

Proof. Full fibers, partial fibers, and empty fibers are mutually exclusive, so the displayed decomposition is canonical. Choose the t exceptional points, the p partial fibers, and the m full fibers in $\binom{b}{t} \binom{N}{p} \binom{N-p}{m}$ ways. On one partial fiber the weight enumerator of the allowed nonempty proper subsets is $(1+x)^c - 1 - x^c$; hence the coefficient in (PO1) counts the choices having total partial weight r . Summing (PO1) and extracting the coefficient of x^a gives

$$[x^a](1+x)^b (1+x^c + ((1+x)^c - 1 - x^c))^N = [x^a](1+x)^{b+cN} = \binom{b+cN}{a}.$$

Multiplying the linear factors belonging to the three disjoint parts of the support proves the locator identity. $\square$

Proposition 8.34 (Partial-occupancy Fourier factorization). *In the setting of Theorem 8.33, let G be a finite abelian group, let $g : D \rightarrow G$, and put $\Phi(S) = \sum_{x \in S} g(x)$. For a character $\chi \in \widehat{G}$ and a regular fiber $F_y = \phi^{-1}(y)$, define*

$$A_y(\chi) = \prod_{x \in F_y} \chi(g(x)), \quad P_y(\chi; z) = \prod_{x \in F_y} (1 + z\chi(g(x))) - 1 - z^c A_y(\chi).$$

Then

$$(PO3) \quad \sum_{S \in \Omega_{t,m,p,r}} \chi(\Phi(S)) = [u^t v^m \zeta^p z^r] \prod_{x \in X} (1 + u\chi(g(x))) \prod_{y \in Q} (1 + vA_y(\chi) + \zeta P_y(\chi; z)).$$

Consequently, for every $s \in G$,

$$(PO4) \quad |\{S \in \Omega_{t,m,p,r} : \Phi(S) = s\}| = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \overline{\chi(s)} \sum_{S \in \Omega_{t,m,p,r}} \chi(\Phi(S)).$$

At the trivial character, (PO3) is exactly the support count (PO1). Thus support add-back is unconditional, while a finite flatness theorem is exactly a bound for the nontrivial-character sum in (PO4).

Proof. For a fixed regular fiber the three terms in the second product encode, respectively, an empty fiber, a full fiber, and a nonempty proper subset. The variables v, ζ, z record the numbers m, p, r , while u records the exceptional weight t . Multiplicativity of χ turns the character of the support sum into the product of its selected-point characters, proving (PO3) by coefficient extraction. Formula (PO4) is character orthogonality on G . $\square$

Remark 8.35 (Effective normalization of a partial-occupancy slice). Fix a nonempty slice $\Omega_\lambda = \Omega_{t,m,p,r}$, choose $S_0 \in \Omega_\lambda$, and let

$$G_\lambda = \langle \Phi(S) - \Phi(S_0) : S \in \Omega_\lambda \rangle \leq G.$$

The attained boundary values lie in the affine coset $\Phi(S_0) + G_\lambda$, and the effective inversion is

$$(PO5) \quad |\{S \in \Omega_\lambda : \Phi(S) = s\}| = \frac{1}{|G_\lambda|} \sum_{\chi \in \widehat{G_\lambda}} \overline{\chi(s - \Phi(S_0))} \sum_{S \in \Omega_\lambda} \chi(\Phi(S) - \Phi(S_0)).$$

Every character of G_λ extends to the ambient finite abelian group, so the inner sum may be evaluated from (PO3) after multiplication by the fixed factor $\chi(\Phi(S_0))$. Different extensions agree on support differences. Equivalently, ambient characters group by their restriction to G_λ , and the annihilator multiplicity cancels exactly against the ambient denominator. This identifies the correct Fourier denominator; it does not assert that the realized image fills the affine group.

Remark 8.36 (What partial-occupancy add-back does and does not prove). The exponentially many local proper-subset choices are already contained inside the coarse slices $\Omega_{t,m,p,r}$; they incur no separate profile-count loss in (PO2). The theorem does not bound the nontrivial Fourier terms in (PO4), the size of a realized boundary image, or the projection of a witness cell to distinct slopes. Those are separate analytic and ray payments.

8.5. Asymptotic and finite regimes.

Theorem 8.37 (Unconditional shallow-prefix RS-MCA exponent). *Let $C_n = \text{RS}_{\mathbb{F}_n}(D_n, k_n)$, where $D_n \subseteq \mathbb{B}_n \subseteq \mathbb{F}_n$, $|D_n| = n$, and $k_n/n \rightarrow \rho \in (0, 1)$. Let $a_n \geq k_n + 1$, put $w_n = a_n - k_n - 1$, and assume*

$$(AS1) \quad \frac{a_n}{n} \rightarrow \alpha \in (0, 1), \quad w_n \log |\mathbb{B}_n| = o(n),$$

$|\mathbb{F}_n| > n$, and, for some finite $\phi \geq 0$,

$$(AS2) \quad \frac{1}{n} \log |\mathbb{F}_n| \rightarrow \phi, \quad \frac{1}{n} \log (|\mathbb{F}_n| - n) \rightarrow \phi.$$

For the full-field challenge,

$$(AS3) \quad \frac{1}{n} \log B_{C_n}^{\text{MCA}}(a_n) = \min\{h(\alpha), \phi\} + o(1),$$

and therefore

$$(AS4) \quad \frac{1}{n} \log e_{\text{MCA}} \left(C_n, 1 - \frac{a_n}{n} \right) = -(\phi - h(\alpha))_+ + o(1).$$

Because $|\mathbb{B}_n| \geq n$, condition (AS1) forces $\alpha = \rho$. No smoothness, Fourier estimate, or auxiliary ray hypothesis is assumed.

Proof. Put $M_n = \binom{n}{a_n}$ and $L_n = \lceil M_n |\mathbb{B}_n|^{-w_n} \rceil$. Exact-agreement reduction and fixed-support slope uniqueness give the unconditional upper bound

$$(AS5) \quad B_{C_n}^{\text{MCA}}(a_n) \leq \min\{|\mathbb{F}_n|, M_n\}.$$

The exact locator-prefix list and collision-aware pole compiler give

$$(AS6) \quad B_{C_n}^{\text{MCA}}(a_n) \geq \left\lceil \frac{L_n(|\mathbb{F}_n| - n)}{|\mathbb{F}_n| - n + k_n(L_n - 1)} \right\rceil.$$

By Stirling's formula and (AS1), $\log L_n = h(\alpha)n + o(n)$. For positive x, y , $xy/(x + k_n y) \geq \frac{1}{2} \min\{y, x/k_n\}$; hence (AS2), $\log k_n = o(n)$, and the ceiling show that the right side of (AS6) has logarithm $n \min\{h(\alpha), \phi\} + o(n)$. The upper bound (AS5) has the same exponent, proving (AS3); subtracting $\log |\mathbb{F}_n|$ proves (AS4). Finally $w_n \log n = o(n)$, so $(a_n - k_n)/n \rightarrow 0$ and $\alpha = \rho$. $\square$

Corollary 8.38 (Unconditional large-challenge RS–MCA threshold). *Under the row hypotheses of Theorem 8.37, let the full-field target satisfy $\varepsilon_n = \exp(-\lambda n + o(n))$ for a finite $\lambda \geq 0$. If*

$$(AS7) \quad \lambda < \phi - h(\rho),$$

then, for every sufficiently large n ,

$$(AS8) \quad a_n^* = k_n + 1, \quad \delta_n^* = 1 - \frac{k_n + 1}{n} = 1 - \rho + o(1).$$

Conversely, if $\lambda > (\phi - h(\rho))_+$, every agreement sequence satisfying $(a_n - k_n - 1) \log |\mathbb{B}_n| = o(n)$ is unsafe. The critical boundary $\lambda = (\phi - h(\rho))_+$ is governed by the exact finite constants and is not decided by exponent asymptotics. This is a large scalar-extension regime. If $|\mathbb{F}_n|$ is polynomial in n , then $\phi = 0$ and (AS7) cannot hold; the corollary does not settle that Proximity-Prize parameter regime.

Proof. At $a = k_n + 1$, (AS5) is $B_{C_n}^{\text{MCA}}(a) \leq \binom{n}{k_n+1} = \exp(h(\rho)n + o(n))$. Under (AS7), $\varepsilon_n |\mathbb{F}_n| = \exp((\phi - \lambda)n + o(n))$ has strictly larger exponent, and taking the integer floor does not change the comparison. Thus the first permitted agreement is safe and (AS8) follows. The converse follows from (AS4) and the strict reverse exponential inequality. $\square$

Corollary 8.39 (Exact binary additive-domain examples). *Let $k = 2^{40}$, let D be any $\mathbb{F}_2$ -linear subspace of the indicated field having the indicated size, and take the target 2^{-128} . The four rows*

k/n	$\mathbb{F}$	n	$\delta_{C, \text{sup}}^*$
1/2	$\mathbb{F}_{2^{166}}$	2^{41}	1/8
1/4	$\mathbb{F}_{2^{168}}$	2^{42}	1/4
1/8	$\mathbb{F}_{2^{169}}$	2^{43}	1/4
1/16	$\mathbb{F}_{2^{170}}$	2^{44}	1/4

have exactly the displayed supremal MCA transitions, and every endpoint is unsafe.

Proof. For $q = 2^m$, the slope budget is $B = \lfloor q 2^{-128} \rfloor = 2^{m-128}$. In the four rows this is respectively $2^{38}, 2^{40}, 2^{41}, 2^{42}$, and $3(B - 1) \leq n - k$. Apply Theorem 1.1 and translate the largest safe grid radius $B - 1$ to the real closed-ball supremum B/n . $\square$

Proposition 8.40 (Exact unsafe test). *Let*

$$L_n = \left\lceil \binom{n}{a_n} |\mathbb{B}_n|^{-(a_n - k_n - 1)} \right\rceil, \quad q_n = |\mathbb{F}_n| > n.$$

If

$$(13.3) \quad \left\lceil \frac{|\Gamma_n|}{q_n} \left\lceil \frac{L_n(q_n - n)}{q_n - n + k_n(L_n - 1)} \right\rceil \right\rceil > B_n^*,$$

then agreement a_n is unsafe at the target. The same statement holds with L_n replaced by any larger identity, quotient, Chebyshev, or remainder-profile list proved for the dimension- $k_n + 1$ code.

Proof. The identity list is [Proposition 7.1](#), and [Theorem 7.2](#) produces a bad-slope set $Z \subseteq \mathbb{F}_n$ of the inner-ceiling size. Replacing a line (r_0, r_1) by $(r_0 + \delta r_1, r_1)$ translates Z by $-\delta$. Averaging over $\delta \in \mathbb{F}_n$ gives $|Z||\Gamma_n|/q_n$ challenge slopes, so some translate has at least the outer-ceiling number. The profile variants use the same conversion after their list sizes have been established. $\square$

Theorem 8.41 (Unconditional exact support-envelope bracket). *Let $C = \text{RS}_{\mathbb{F}}(D, k)$, let $D \subseteq \mathbb{B} \subseteq \mathbb{F}$, put $q = |\mathbb{F}| > n$, fix $\emptyset \neq \Gamma \subseteq \mathbb{F}$, and let $B^* = \lfloor \varepsilon |\Gamma| \rfloor$. For $k + 1 \leq a \leq n$, define*

$$(SB1) \quad \begin{aligned} L(a) &= \left\lceil \binom{n}{a} |\mathbb{B}|^{-(a - k - 1)} \right\rceil, \\ P(a) &= \left\lceil \frac{|\Gamma|}{q} \left\lceil \frac{L(a)(q - n)}{q - n + k(L(a) - 1)} \right\rceil \right\rceil, \\ U(a) &= \min \left\{ |\Gamma|, \binom{n}{a} \right\}. \end{aligned}$$

If $a_- < a_+$ satisfy

$$(SB2) \quad P(a_-) > B^*, \quad U(a_+) \leq B^*,$$

then the global target threshold exists and

$$(SB3) \quad a_- < a^* \leq a_+.$$

If $a_+ = a_- + 1$, then $a^ = a_+$ exactly. In particular,*

$$(SB4) \quad \binom{n}{k+1} \leq B^* \implies a^* = k + 1.$$

These are unconditional finite comparisons; (SB2) is the literal target reserve and contains no asymptotic placeholder.

Proof. The exact prefix list and pole-line construction give $B_{C,\Gamma}^{\text{MCA}}(a) \geq P(a)$ by [Proposition 8.40](#). The exact support atlas gives $B_{C,\Gamma}^{\text{MCA}}(a) \leq U(a)$ by [Proposition 2.11](#). Thus a_- is unsafe and a_+ is safe; monotonicity proves (SB3) and the adjacent conclusion. For (SB4), the first permitted agreement is already safe. $\square$

9. CONCLUSION

The finite content of the paper is the quadratic mean-overlap theorem: a tangent construction supplies $r + 1$ slopes, and a line-secant overlap argument proves that no received line supplies more. We are unaware of a prior finite MDS theorem of this form, and its small parameter pockets are not certified by the specialized estimate ([LC5](#)). The example $(n, k, r) = (32, 15, 6)$, propagated to positive-density smooth and circle prime families, isolates this finite contribution.

At constant relative radius the prior BCHKS CA theorem, combined with the below-half equality of ACFY/WHIR, already gives the stronger global conclusion; the exact sparse decomposition here supplies a self-contained and challenge-restricted route. Making the literature implication integral completes the exact staircase through every compact subinterval below half the minimum distance and widens the target compiler correspondingly. The circle statements

add no hidden coding assumption: monomial and stereographic equivalences preserve each bad-slope set, not only its cardinality.

The target comparison remains genuinely discrete. If $b = \lfloor \varepsilon |\Gamma| \rfloor$, the largest safe grid point is $(b - 1)/n$, while the real-radius safe set can be half open with unsafe supremum b/n . The certified Proth and Mersenne-31 rows illustrate this endpoint bookkeeping, but the four Proth thresholds are not claimed as new consequences.

No theorem here reaches the unrestricted near-capacity smooth/circle frontier. The fixed-length prime families have positive density only with length held fixed, and growing dyadic prime-field families necessarily have density zero. Progress beyond half distance still requires new received-line or list-decoding information; no computational certificate, profile ledger, or unproved equidistribution statement has been promoted to such an input in this paper.

APPENDIX A. VERIFICATION DETAILS

This appendix records the elementary coding-theoretic and integer calculations used above in a form that can be checked independently of the geometric proofs. Some of these facts are standard, but including their short derivations fixes all normalization conventions and makes the four large finite rows reproducible from the decimal data printed in the paper.

A.1. Reed–Solomon and MDS conventions. Let $D = \{x_1, \dots, x_n\} \subseteq \mathbb{F}$ consist of distinct points and let $1 \leq k < n$. Evaluation on D is injective on $\mathbb{F}[X]_{<k}$: a nonzero polynomial of degree below k cannot vanish at the $n \geq k$ distinct points of D . It follows that $\text{RS}_{\mathbb{F}}(D, k)$ has dimension k . If two codewords are distinct, their difference is the evaluation of a nonzero polynomial of degree at most $k - 1$, hence has at most $k - 1$ zero coordinates and at least $n - k + 1$ nonzero coordinates. Conversely, the evaluation of $\prod_{i=1}^{k-1} (X - x_i)$ has exactly $k - 1$ zero coordinates. Thus the minimum distance is

$$d_{\min} = n - k + 1.$$

This proves directly that the code is MDS.

For $x \in D$, put

$$\lambda_x = \left(\prod_{y \in D \setminus \{x\}} (x - y) \right)^{-1}.$$

The Lagrange interpolation formula for a polynomial f of degree at most $n - 1$ shows that its leading coefficient is $\sum_{x \in D} \lambda_x f(x)$. Applying this to $f(X) = X^j$, $0 \leq j \leq n - 2$, gives

$$\sum_{x \in D} \lambda_x x^j = 0.$$

Let $R = n - k$ and let H be the R -by- n matrix whose column at x is

$$h_x = \lambda_x (1, x, \dots, x^{R-1})^\top.$$

If $P \in \mathbb{F}[X]_{<k}$, then the j -th entry of $H(P(x))_{x \in D}^\top$ is $\sum_x \lambda_x x^j P(x) = 0$, because $j + \deg P \leq n - 2$. Thus H annihilates the code. Every R -by- R submatrix of H is a nonzero diagonal scaling of a Vandermonde matrix, so it is invertible. Hence H has rank R , its kernel has dimension k , and its kernel is exactly $\text{RS}_{\mathbb{F}}(D, k)$.

The same Vandermonde calculation proves that every set of at most R columns is linearly independent. In particular, for $E \subseteq D$, $|E| \leq R$, the space $V_E = \text{Span}\{h_x : x \in E\}$ has dimension $|E|$. A word u agrees with a codeword outside E if and only if its syndrome lies in V_E . Indeed, writing $u = c + e$ with e supported on E gives one direction. Conversely, if $Hu^\top = \sum_{x \in E} e_x h_x$, subtracting the word whose coordinates on E are e_x leaves zero syndrome and therefore a codeword. This is the precise error-support form of the syndrome normalization used in [Proposition 3.1](#).

For completeness, the fixed-support slope uniqueness used repeatedly in the paper is also an MDS-free linear observation. Suppose a support S explains $u_0 + \gamma u_1$ and $u_0 + \gamma' u_1$ for distinct γ, γ' . Subtracting the two explaining codewords and dividing by $\gamma - \gamma'$ explains u_1 on

S ; subtracting γ times this explanation from the first explains u_0 there. Thus S is common. Consequently a noncommon support contributes at most one finite MCA-bad slope.

Lemma A.1 (Monotonicity and exact endpoint conversion). *For fixed C and Γ , the integer numerator $B_{C,\Gamma}^{\text{MCA}}(a)$ is nonincreasing in a . Let $B^* = \lfloor \varepsilon |\Gamma| \rfloor$, and suppose a_0 is the first safe agreement, while $a_0 - 1$ is unsafe. Then the largest safe grid radius is $1 - a_0/n$, whereas the safe real radii form*

$$\left[0, \frac{n - a_0 + 1}{n}\right),$$

so their supremum is one grid step larger and is not attained.

Proof. Every witness at agreement at least $a+1$ is also a witness at agreement at least a , proving monotonicity. For a real radius δ , the closed Hamming ball permits precisely $r = \lfloor \delta n \rfloor$ errors and therefore requires agreement $n - r$. The condition $n - \lfloor \delta n \rfloor \geq a_0$ is equivalent to $\lfloor \delta n \rfloor \leq n - a_0$, or $\delta < (n - a_0 + 1)/n$. The largest grid point in that interval is $(n - a_0)/n = 1 - a_0/n$. $\square$

This lemma explains the half-open intervals in the main theorem. In particular, at target 2^{-128} , writing $B = \lfloor q/2^{128} \rfloor$, exact numerator B at error radius $B - 1$ and numerator at least $B + 1$ at radius B give the safe set $[0, B/n)$, rather than a closed interval.

A.2. The quadratic boundary. Put $R = n - k$ and

$$F_{n,k}(r) = r^2 - n(3r - R).$$

The mean-overlap hypothesis is $F_{n,k}(r) \geq 0$. Its two roots are

$$\frac{3n \pm \sqrt{9n^2 - 4nR}}{2} = \frac{3n \pm \sqrt{n(5n + 4k)}}{2}.$$

Since $F_{n,k}(0) = nR > 0$ and $F'_{n,k}(r) = 2r - 3n < 0$ for $0 \leq r \leq n$, the admissible initial interval ends at

$$r_{\text{quad}}(n, k) = \left\lfloor \frac{3n - \sqrt{n(5n + 4k)}}{2} \right\rfloor.$$

If $k/n \rightarrow \rho$, division by n gives

$$\frac{r_{\text{quad}}(n, k)}{n} \rightarrow \alpha_\rho = \frac{3 - \sqrt{5 + 4\rho}}{2}.$$

The derivative observation also shows that checking $F_{n,k}(B - 1) \geq 0 > F_{n,k}(B)$ proves the exact identity $r_{\text{quad}}(n, k) = B - 1$; no floating-point approximation is involved.

At the four challenge rates the limiting constants are, for orientation,

ρ	α_ρ	decimal value
1/2	$(3 - \sqrt{7})/2$	0.1771243444...
1/4	$(3 - \sqrt{6})/2$	0.2752551286...
1/8	$(6 - \sqrt{22})/4$	0.3273960600...
1/16	$(6 - \sqrt{21})/4$	0.3543560762...

The decimals are not used in any proof; the exact signs in (PC0) locate the finite boundaries.

A.3. Proth certificates and budget divisions. We spell out the certificate format. Suppose $p = u2^s + 1$, where u is odd and $u < 2^s$, and suppose an integer a_0 satisfies

$$a_0^{(p-1)/2} \equiv -1 \pmod{p}.$$

If a prime ℓ divides p , then the order of a_0 modulo ℓ has exact 2-part 2^s . Hence $2^s \mid \ell - 1$, so $\ell \geq 2^s + 1$. The inequality $u < 2^s$ gives $p = u2^s + 1 < 2^{2s} + 1$, and therefore $2^s > \sqrt{p} - 1$. If p were composite it would have a prime divisor at most $\sqrt{p}$, contradicting the lower bound $\ell \geq 2^s + 1 > \sqrt{p}$. Thus p is prime. This is Proth's theorem in exactly the form needed here.

All certificate inputs are collected below. The equality in the last column is a residue equality modulo the corresponding p .

ρ	s	u	a_0	bits(p)	$a_0^{(p-1)/2} \bmod p$
1/2	92	26766274163673319604503	3	167	$p - 1$
1/4	93	41595378994516821279015	13	169	$p - 1$
1/8	95	24737346889219389259839	5	170	$p - 1$
1/16	97	13387194060291799253121	5	171	$p - 1$

Together with the decimal values of p printed before [Proposition 6.1](#), this table is a complete Proth certificate: the reader checks the single exact identity $p - 1 = u2^s$, the parity and size of u , and one modular exponentiation. Modular exponentiation is an integer operation and does not rely on a probabilistic primality test.

For a completely mechanical check, first compute $t = a_0^u \bmod p$ by binary square-and-multiply, then square t exactly $s - 1$ times modulo p . The certificate accepts precisely when the result is $p - 1$. The algorithm uses $O(\log u + s)$ modular multiplications. The remaining row checks use only division with remainder. Writing

$$p = B2^{128} + r_p,$$

the four exact remainders are

ρ	r_p
1/2	1381541083842484386787422633985
1/4	2921538492713497448761933168641
1/8	2495687119199326634196634435585
1/16	20440865928680199099134339186689

and each lies strictly between 0 and $2^{128} = 340282366920938463463374607431768211456$. Therefore the displayed B is exactly $\lfloor p/2^{128} \rfloor$.

Finally, $n = 2^e$ with $e = 41, 42, 43, 44$, respectively, while $s = 92, 93, 95, 97$. Thus $n \mid 2^s \mid p - 1$, and the cyclic group $\mathbb{F}_p^\times$ contains a unique subgroup of order n . Each field has the printed bit length, all smaller than 256, and $k = 2^{40}$ gives the required four rates. The exact boundary signs are

ρ	$F_{n,k}(B - 1)$	$F_{n,k}(B)$
1/2	5154112775168	-663955886271
1/4	7590647904465	-3182321912768
1/8	13908181940112	-6720484728007
1/16	19335616403905	-20973145690236

and hence $B - 1 = r_{\text{quad}}(n, k)$ in every row.

Proposition A.2 (Independent certificate verification). *The data in [Table 1](#) define four prime fields, four power-of-two multiplicative subgroups of the stated lengths, and slope budgets $B = \lfloor p2^{-128} \rfloor$. In every row the exact safe set is $[0, B/n)$, and the endpoint is unsafe.*

Proof. The Proth calculation proves primality, the divisibility calculation produces the subgroup, and the remainder calculation gives the target budget. The two signs locate the quadratic boundary at $B - 1$. Apply [Corollary 4.12](#) and then [Lemma A.1](#). Every step is an equality or inequality between the printed integers. $\square$

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